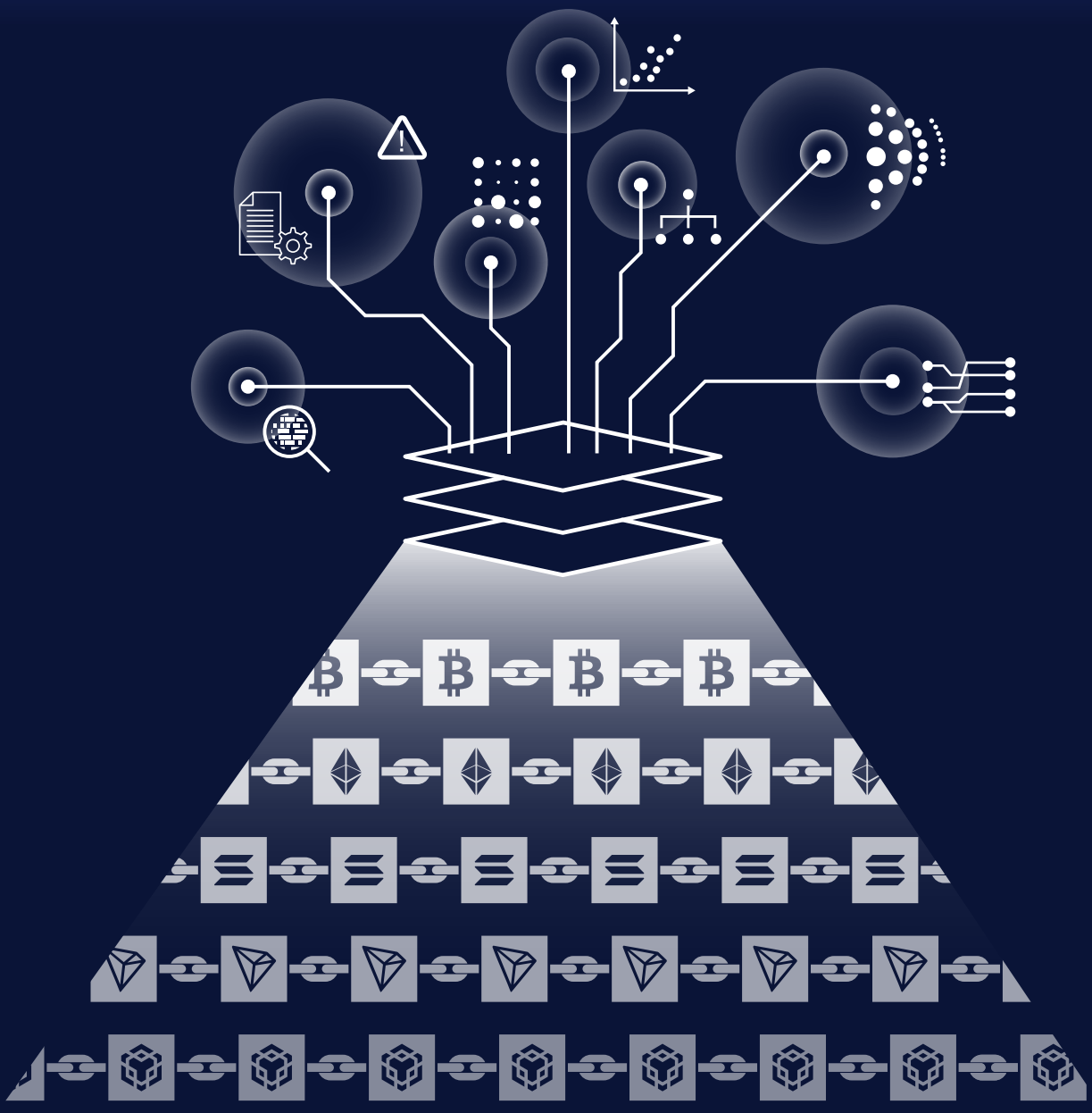
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Joint Research Centre

CIPHER: Monitoring crypto markets

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Abstract

This progress report presents the current state of development of the **CIPHER framework**, a robust and scalable platform for monitoring crypto-asset markets, decentralized finance (DeFi) protocols, and blockchain-based financial activity. CIPHER's role is to integrate blockchain-derived data with structured monitoring modules, covering areas such as **market activity, financial stability risks, and systemic dependencies**.

The report outlines the methodological foundations, technical infrastructure, and preliminary analytical outputs achieved to date. All of this is contextualised within the recent **technological and regulatory development of the crypto landscape**.

The report highlights key **challenges and lessons learned**, including data issues, the computational demands of large-scale blockchain analytics, and the need for interoperability and long-term sustainability. Finally, it provides a forward-looking **roadmap for further development**, including infrastructure scaling, feature enhancements, and deeper collaboration with EU and international stakeholders.

The ultimate objective of CIPHER is to provide a **policy-relevant, operationally sustainable monitoring and analytics tool** capable of supporting EU institutions in supervisory, regulatory, and strategic decision-making related to crypto-assets and DeFi.

Executive summary

Crypto-assets and decentralized finance (DeFi) are no longer a niche innovation but a rapidly expanding system with direct implications for **Europe's sovereignty and financial security**. With the market capitalisation of crypto assets **growing in the last two years by 400 %** to about **USD 4 trillion in September 2025**, the question for the EU is not whether these markets matter anymore, but rather how to measure, monitor, and respond to them in a way that safeguards financial stability, monetary autonomy, and consumer protection. Understanding these dynamics requires tools capable of converting vast, complex, and pseudonymous blockchain data into actionable insights.

The **CIPHER project** addresses this challenge. It continuously ingests **live data** from the Bitcoin and Ethereum blockchains, which together account for **approximately 80% of global crypto assets**, and converts this massive, pseudonymous data stream into policy-relevant intelligence. Built on layers of data import, storage, analytics, and interactive dashboards, CIPHER makes it possible for EU financial stability authorities, regulators, and supervisors to track market activity, assess systemic dependencies, and anticipate emerging risks both in **technical detail** and **high-level overviews**. Under a unified, single framework, CIPHER already allows users to seamlessly work with **terabytes of data** and **billions of transactions.**

Yet, as innovation in the crypto sector accelerates, new challenges and opportunities are emerging. Tokenisation of real-world assets such as sovereign bonds and real estate is expanding (Section 2.2.1), and DeFi protocols are **replicating traditional financial services without intermediaries**. Other innovations aim to make blockchains faster and cheaper, though they can also make oversight more difficult (Section 2.2.2). Tools that connect different blockchains are becoming central to the system but have already proven vulnerable to attacks (Section 2.2.3), while decentralized decision-making is expanding without a clear legal accountability framework (Section 2.2.4). The **global and experimental nature of DeFi underscores the need for technological monitoring and innovative policy approaches** to address emerging risks, as the landscape presents **significant oversight gaps**.

Regulatory frameworks struggle to keep up and remain **fragmented** across jurisdictions. The EU's Markets in Crypto Assets (MiCA) regulation provides licensing and market integrity rules, yet **enforcement against decentralized or offshore actors remains limited** (Section 2.1.2).

CIPHER allows these developments to be observed in real time. For instance, when the **U.S.** GENIUS Act **of 2025** strengthened the role of dollar-denominated stablecoins, CIPHER showed a **surge in the volume** of the USDe stablecoin, illustrating how regulation abroad can reshape global liquidity patterns (Section 3.2.3). In fact, **99.6% of the current stablecoin market capitalization is linked to the U.S. dollar** and backed in part by U.S. Treasury bonds[1], deeply anchoring the crypto ecosystem to U.S. financial markets while leaving the euro's role marginal. Such dependencies have

[1] Current market capitalization of all stablecoins as of September 2025 (in US dollars) is 287 billion. Out of this, an overwhelming 286 billion (99.6%) is pegged to the US dollar, 460 million (0.16%) pegged to the Russian rubble in the second place and 240 million (0.08%) pegged to the euro in the third place. See https://defillama.com/stablecoins for more info.

the potential to **directly affect the future of EU's monetary sovereignty** (Sections 2.3.4 and 3.3.3).

Risks are not confined to regulatory events though, but also to market dynamics. CIPHER in fact shows how perceived **risk to stablecoin stability can ripple across the system**, as shown by the historical case of the Silicon Valley Bank failure (Section 3.2.3). Along with other similar cases, this event highlights that power is often **concentrated** in the hands of a few players (Section 2.3.1), even if this goes against the original idea of decentralisation and thus creates weak points in the system.

Meanwhile, **links to traditional finance are tightening:** banks provide custody, hedge funds increase exposure, and stablecoins connect directly to sovereign debt markets (Section 2.3.2). Combined with the sector's notorious **volatility**, these interconnections heighten the potential for shocks to **spill over** into the wider economy, while new technologies like artificial intelligence could make these effects even more pronounced (Section 2.3.3).

Beyond monitoring, CIPHER is conceived as a **policy-relevant laboratory for scenario analysis** (Section 3.3.2). Its stress-testing modules are envisaged to simulate shocks - such as de-peg events in major stablecoins or failures of top protocols - directly on blockchain data, showing how crises could propagate across markets. Leveraging the native crypto infrastructure for running a hypothetical simulation of the effect of the stress on the market, this approach has the potential to **support systemic risk analysis at the ECB, ESMA, and national supervisors**.

CIPHER will continue to evolve as a **scalable, sustainable and adaptable platform**. Priorities for the short term include expanding data coverage, integrating more blockchains and off-chain data sources, and building DeFi stress-testing capabilities to generate forward-looking insights into potential risk propagation.

These efforts will go hand in hand with the expansion of **computational capacities,** improvements in indexing, storage, and adjustments in the data processing pipelines. In doing so, it will provide the EU with a unique capacity to detect risks early, assess the strategic implications of the increasing crypto's global adoption, and evaluate the impact of regulatory choices.

Further **collaboration** with EU institutions and international partners is expected to facilitate shared analyses, benchmarking, and convergence on monitoring standards, thereby reinforcing the evidence base for policy and regulatory decision-making. Hosted at the European Commission's **Joint Research Centre** (JRC), CIPHER benefits from the JRC's interdisciplinary expertise and technical orientation pose a unique advantage for this type of platform, positioning it as a natural hub to connect advanced blockchain analytics with policy needs. Through cooperation with EU institutions and international partners, CIPHER's long-term vision is to strengthen the EU's ability to protect financial stability and sovereignty in the rapidly evolving landscape of digital finance.

1. Introduction

This document serves as a comprehensive update on the current state of the **CIPHER**[2] (Crypto Intelligence Platform for a Holistic European Regulation) project, **a monitoring and analytical framework** developed to address the complexities of the rapidly evolving cryptocurrency and decentralized finance (DeFi) ecosystem. As digital assets become increasingly integrated into traditional financial systems, the need for robust, data-driven tools to support regulatory oversight and policy-making has grown critical. CIPHER strives to meet this demand by **enriching** blockchain **data with additional information, performing complex analytics, and allowing real-time visualization**. This combination of tools is designed to **deliver actionable insights** for policymakers, regulators, and stakeholders.

This report provides a detailed overview of the project's technical infrastructure, current capabilities, and early use cases, while also addressing the challenges encountered during development and outlining the roadmap for future enhancements. Structured into two key chapters:

- Chapter 2 contextualizes the **evolving crypto landscape**, highlighting regulatory trends, technological advancements, and emerging risks.
- Chapter 3 details **CIPHER's architecture**, including its data import, storage, analytics, and visualization layers. It also explores the project's **practical applications**, such as stablecoin monitoring and DeFi risk assessment with concrete examples. Finally, the chapter includes **lessons learned and next steps**, ensuring alignment with the EU's priorities for innovation, financial stability, and strategic autonomy.

By focusing on the project's progress to date, this report aims to inform stakeholders about CIPHER's role in addressing the dual imperatives of **fostering innovation and safeguarding the integrity of the financial ecosystem**. It underscores how the framework is being operationalized to **support policy workflows, financial stability monitoring, and cross-border regulatory coordination**, while also flagging the technical and strategic hurdles that remain.

This milestone reflects the JRC commitment to developing a **data-driven, adaptive approach** to digital finance governance, ensuring that CIPHER evolves in tandem with the dynamic crypto landscape. The insights and findings presented here are expected to guide future development, stakeholder engagement, and the integration of CIPHER into broader regulatory and policy frameworks.

[2] CIPHER is a JRC exploratory project expected to run for three years. This report represents the first update on the progress of the project after its first year.

2. Evolving crypto landscape: Regulation, market dynamics and strategic risks

The global crypto and DeFi ecosystem has entered a period of **rapid transformation**. New technologies, shifting regulatory stances, and the increasing integration of digital assets with traditional financial systems are creating both **opportunities and risks** for policymakers. For the European Union, these developments intersect with key priorities, including financial stability, anti-money laundering, consumer protection, and strategic autonomy.

CIPHER is being developed against this backdrop to provide the **monitoring, analytical, and horizon-scanning capabilities** needed to navigate such a fast-changing and globally interconnected environment. This section provides insight in this respect on **regulatory trends, technological developments, risks and potential disruption scenarios**.

Readers already familiar with the crypto and DeFi ecosystem may skip directly to Chapter 3, which details the CIPHER platform and provides examples of applications and potential avenues for future development.

2.1. Regulatory and policy trends

Many advances in regulation have recently happened at both the EU as well as the international level. The EU's MiCA and the US' GENIUS Act are just two examples of approaches policymakers might take when faced with the complex world of crypto and DeFi.

2.1.1. Global development

Building on innovations in digital assets and governance, **regulatory approaches are diverging sharply across jurisdictions**, with profound implications for market structure and risk.

In July 2025, the **U.S. enacted the GENIUS Act**[3], the first federal law specifically governing stablecoins. It mandates full one-to-one backing with high-quality liquid assets such as USD and Treasury bills, monthly reserve disclosures, and independent audits. Complemented by the CLARITY Act and an executive decision against a U.S. Central Bank Digital Currency[4], these measures aim to **entrench the U.S. dollar's**

Global Development

Why it matters:

Without competitive euro-denominated stablecoins and a supportive regulatory posture, the EU risks **ceding market influence** and governance over token-based monetary systems to foreign jurisdictions.

[3] See in the Financial Times at https://www.ft.com/content/4c41e6e8-374c-4b1a-ac7f-88c245fb18c8, in Reuters at https://www.reuters.com/legal/government/trump-signs-stablecoin-law-crypto-industry-aims-mainstream-adoption-2025-07-18/, or Wikipedia: https://en.wikipedia.org/wiki/GENIUS_Act

[4] See for instance The Kiplinger, https://www.kiplinger.com/investing/cryptocurrency/genius-clarity-anti-cbdc-acts-what-bitcoin-investors-need-to-know, the same news covered by crypto-oriented media at https://thecurrencyanalytics.com/altcoins/ripple-supports-new-white-house-crypto-strategy-targeting-regulatory-clarity-and-innovation-188129, or Wikipedia: https://en.wikipedia.org/wiki/Executive_Order_14178

dominance in tokenized finance while excluding central bank–issued digital alternatives.

By contrast, several Asian and offshore financial hubs continue to attract crypto firms through comparatively **innovation-friendly regulatory frameworks**. Jurisdictions such as Singapore, Hong Kong, and the Cayman Islands offer licensing regimes that are often viewed as more flexible than those in the US or EU. Features such as lower tax burdens, adaptable corporate structures, and regulatory pathways designed to accommodate new business models have encouraged cross-border migration of firms and liquidity toward these environments. This divergence **creates asymmetries in compliance costs, supervisory reach, and risk exposure**, while also complicating coordinated oversight of global stablecoins, cross-chain transactions, and DeFi protocols.

2.1.2. MiCA Implementation

To keep up with the global development, The Markets in Crypto-Assets Regulation (MiCA) marks the EU's most ambitious attempt to legislate in the crypto-asset space. Officially adopted in May 2023 and fully applicable from **30 December 2024**, MiCA imposes requirements on asset-referenced tokens (e.g., stablecoins) and crypto-asset service providers (CASPs), including licensing, reserve transparency, market abuse controls, and white-paper disclosures[5].

Despite these advances, significant **enforcement challenges remain**. Many regulated actors span multiple jurisdictions, operate partially offshore, or function through decentralized infrastructures. For instance, a DeFi lending protocol running on immutable Ethereum smart contracts may **serve EU users without any EU-based legal entity**, making MiCA's enforcement practically limited.

MiCA

Why it matters:

Without enhanced tracking and monitoring, the legal clarity **MiCA** provides may not translate into **effective oversight**.

Proper understanding and monitoring of the underlying protocols and entities is needed to **ensure compliance**.

The 2022 **sanctioning of Tornado Cash[6]** by the U.S. Treasury exemplifies these limitations. Despite formal sanctions, the protocol continues to operate on decentralized networks, circumventing traditional enforcement mechanisms[7]. **Regulatory fragmentation** across

[5] Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets, and amending Regulations (EU) No 1093/2010 and (EU) No 1095/2010 and Directives 2013/36/EU and (EU) 2019/1937 https://eur-lex.europa.eu/eli/reg/2023/1114/oj/eng

[6] Tornado Cash was sanctioned by the U.S. Treasury's Office of Foreign Assets Control (OFAC) in August 2022 due to its role in facilitating large-scale money laundering and sanctions evasion. The protocol, a decentralized mixing service built on Ethereum, enables users to obscure the origin and destination of funds by pooling and redistributing transactions. According to U.S. authorities, Tornado Cash was used to launder over USD 7 billion in virtual assets since its creation, including significant amounts linked to cybercrime groups such as the Lazarus Group, which is associated with North Korea's state-sponsored hacking activities.

[7] As a mitigating factor, centralized exchanges (CEXes) generally refuse to accept deposits or transfers that have interacted with Tornado Cash, limiting its accessibility through regulated financial channels. While this reduces

jurisdictions creates **enforcement arbitrage** opportunities, enabling illicit actors to exploit less regulated environments or employ privacy-enhancing techniques to evade detection. The absence of harmonized global regulatory standards complicates coordinated enforcement.

2.1.3. Anti-Money Laundering and Traceability

MiCA complements the EU's broader Anti Money Laundering package, which extends the "travel rule" to crypto transfers. meaning that whenever crypto is transferred, information about both the sender and the recipient must go with the transaction.

Despite this, tools that hide or obscure crypto transactions persist. Already mentioned mixers, like Tornado Cash and privacy coins such as Monero and Zcash remain semi-accessible and can be used to move illicit flows, often evading detection despite sanctions[8]. As a result of this, major crypto exchanges have de-listed privacy coins[9] and the EU is considering their ban[10], while the Tornado cash mixer has been banned by the US Treasury.

Some services have been shut down in the past, but open-source code allows near-instant redeployments, undermining traceability. Constant technological and market research is crucial for keeping up with the increasingly sophisticated money laundering techniques.

Anti-Money Laundering

Why it matters:

Persistent privacy tools and decentralized passageways render **AML** frameworks **less effective** unless supported by **granular on-chain analytics**.

Constant market and technological research are vital.

2.1.4. Remaining DeFi regulatory oversight gaps

While global regulatory initiatives such as MiCA and the GENIUS Act have advanced oversight of crypto-assets, significant **blind spots remain**, particularly for decentralized finance (DeFi) protocols that operate across borders and outside traditional regulatory frameworks. The following section examines these persistent gaps and the challenges they pose for effective supervision.

Irrespective of the physical location, DeFi protocols now **replicate and extend many functions of the traditional financial system**, including lending, trading, derivatives, and automated market making. They do so **without identifiable intermediaries** subject to existing prudential or licensing regimes, making individual regulation on a state - or country - level inherently difficult.

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mainstream use, it does not prevent the movement of funds entirely. "Tainted" funds—those that have passed through Tornado Cash—can still be sent directly to individuals, including influencers or politicians, circumventing regulatory oversight. This poses reputational and legal risks, as recipients may unknowingly receive illicit or sanctioned funds, creating challenges for enforcement and accountability.

[8] See https://dastcoins.com/2025/07/21/privacy-coins-face-crackdown-will-monero-and-zcash-survive/

[9] See https://www.binance.com/en/square/post/585415

[10] See https://coinedition.com/eus-new-legislation-to-ban-monero-zcash-and-non-kyc-wallets/

Governance is typically exercised through on-chain token voting, but token ownership is often highly concentrated. In practice, **effective control may rest with a small number of actors,** including project founders, venture capital investors, or core developers, undermining the notion of decentralisation and raising questions of accountability.

Oversight challenges stem from multiple dimensions. Anonymity and pseudonymity complicate the attribution of decision-making power and the enforcement of rules such as AML obligations or investor protection. Smart contracts execute transactions automatically, which **reduces operational risk** but also **removes conventional points of intervention** where regulators would normally exercise oversight. Furthermore, DeFi protocols can be **deployed globally** and **accessed without geographic restrictions**, which blurs jurisdictional boundaries and risks **regulatory arbitrage**.

DeFi oversight gaps

Why it matters:

The **uncertainty in legal accountability** within **DeFi** undermines supervisory capability. As a result, users can face losses without resource or transparency.

Effective control rests with a small number of pseudonymous private actors, raising questions about the actual level of **decentralization** and the associated **accountability**.

In the EU context, **DeFi is not explicitly covered by** MiCA. The European Securities and Markets Authority (ESMA) and the European Banking Authority (EBA) have both **flagged DeFi as a blind spot**[11] in the regulatory perimeter, especially regarding investor protection and systemic risk. Globally, the Financial Stability Board (FSB) in its 2022–2023 reports[12] has stressed that DeFi activities often mirror those of traditional finance while **lacking equivalent safeguards**, highlighting **risks from leverage, liquidity mismatches, and interconnectedness**. The Bank for International Settlements (BIS) has similarly cautioned that DeFi's governance structures may **concentrate power** among insiders despite claims of decentralisation[13].

These gaps have direct implications for financial stability and consumer protection. Protocols can reach systemic scale quickly, with assets locked in the tens of billions of euros[14], yet their **resilience to stress remains untested**.

[11] European Banking Authority and European Securities and Markets Authority. (2025). Joint report on recent developments in crypto assets. https://www.eba.europa.eu/publications-and-media/press-releases/eba-and-esma-analyse-recent-developments-crypto-assets

[12] Financial Stability Board. (2023). The financial stability implications of decentralized finance. https://www.fsb.org/2023/02/the-financial-stability-risks-of-decentralised-finance/

[13] Aramonte S. et al. (2021) Bank for International Settlements. DeFi risks and the decentralization illusion. https://www.bis.org/publ/qtrpdf/r_qt2112b.htm

[14] See statistics on total value locked for major DeFi protocols at DefiLlama: https://defillama.com/protocol/lido, or https://defillama.com/protocol/aave, or https://defillama.com/protocol/eigenlayer as well as many others.

Interestingly, the collapse of Terra/Luna[15] in 2022 showed how a crypto crash, in this case triggered by the failure of an algorithmic stablecoin, could hit major firms and thus **cascade** rapidly across the ecosystem. .

Finally, the dynamic and experimental nature of DeFi creates a moving regulatory target. Governance models evolve, protocols can fork or migrate liquidity overnight, and new layers such as cross-chain bridges or algorithmic stablecoins introduce additional risks. This combination of **high complexity, rapid innovation, and weak accountability** makes DeFi particularly **difficult to supervise** within the current regulatory perimeter. Addressing DeFi oversight will likely require both technological monitoring tools as well as novel policy approaches.

2.2. Industry and technology development

While the regulatory and policy sphere is trying to catch up with the recent crypto and DeFi innovations, additional technological and industry applications are rapidly expanding within this landscape. These include for instance the tokenization of real-world assets, additional layers on top of basic blockchains, cross-chain bridges or decentralized governance innovations.

2.2.1. Tokenization of real-world assets (RWAs)

Real-world assets (RWAs) refer to traditional, off-chain assets, such as government bonds, commodities, or real estate, that are **digitized** and represented as tokens on a blockchain. Unlike purely digital crypto-assets (e.g., Bitcoin or Ether), RWA tokens are **backed by claims** on tangible or financial instruments in the traditional economy. In practice, this involves a legal and technical bridge: a custodian or issuer holds the underlying asset and creates a corresponding token that can be traded, transferred, or used in decentralized applications.

The aim is to bring the **efficiency, programmability, and global reach** of blockchain infrastructure to conventional asset classes, enabling faster settlement, fractional ownership, and broader access for investors. These tokenized assets are experiencing rapid growth. Notably, this market grew from approximately $5 billion in 2022 to over $24 billion by June 2025, marking a **380 percent increase within the past three years**[16].

Tokenization, RWAs

Why it matters:

The **tokenization** of traditional financial instruments within blockchain ecosystems creates **hybrid value flows**, demanding regulators monitor both crypto-native and legacy asset interactions.

[15] Liu, J. et al. (2023). Anatomy of a Run: The Terra Luna Crash. *NBER Working Paper No. w31160.* https://www.nber.org/papers/w31160

[16] See Redstone finance https://blog.redstone.finance/2025/06/26/real-world-assets-in-onchain-finance-report/ for further reference.

Institutional platforms such as Paxos Labs[17] are **facilitating the embedding of tokenized assets** into DeFi protocols, **linking** traditionally safe and highly liquid assets (e.g. U.S. Treasuries) to the volatility of crypto markets and introducing **both opportunities and risks**. On one hand, RWA tokenization may enhance **market efficiency**, improve **settlement speed**, and expand **investor access** to traditionally illiquid assets. On the other hand, it raises questions of **custody, investor protection, and systemic contagion**, particularly when for example tokenized sovereign debt becomes a core building block of DeFi ecosystems.[18] [19] [20] [21]

Tokenized U.S. Treasury bills surged from about $100 million in January 2023 to roughly $7.5 billion by mid-2025, with BlackRock's BUIDL product capturing 40 percent market share. Commodities tokenization, particularly gold, grew to $1.6 billion by June 2025, led by Paxos Gold (PAXG) and Tether Gold (XAUT)[22].

From a policy and regulatory perspective, tokenization of RWAs is gaining increasing attention. The European Central Bank (ECB) has highlighted tokenization as a **transformative trend** that could **reshape securities markets**, noting both its potential to streamline settlement and its risks if market infrastructures fragment. In the EU, the Pilot Regime on Distributed Ledger Technology (DLT), in force since March 2023, provides a temporary regulatory framework for tokenized financial instruments, allowing experimentation while maintaining safeguards. In addition, the European Securities and Markets Authority (ESMA) has also provided further guidelines on classification of crypto assets as financial instruments to ensure investor protection and orderly market functioning. Globally, the **Financial Stability Board (FSB)** is assessing implications of tokenization for cross-border markets, with specific concern about **legal enforceability of digital claims**, operational resilience of custodians, and risks of interlinkages between **traditional and decentralized finance**.

The BIS has warned that widespread use of tokenized RWAs in DeFi could amplify shocks[23], as traditionally **stable assets become exposed** to the leverage, liquidity mismatches, and governance vulnerabilities of decentralized platforms.

In this context, while RWA tokenization is often presented as a **bridge** between traditional finance and DeFi, it may in fact represent a **new channel of systemic risk** transmission. Supervisors face fundamental questions: who is the accountable issuer or custodian, how ownership rights are legally enforced across jurisdictions, and what safeguards exist in the event of a protocol failure or

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[17] See Paxos Labs: https://paxoslabs.com/

[18] Cipollone, P. (2024). Tokenisation of financial instruments and central bank money settlement. European Central Bank

[19] European Securities and Markets Authority. (2023). DLT Pilot Regime. https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/dlt-pilot-regime

[20] European Securities and Markets Authority. (2025). Guidelines on the conditions and criteria for the qualification of crypto assets as financial instruments. https://www.esma.europa.eu/document/guidelines-conditions-and-criteria-qualification-crypto-assets-financial-instruments

[21] Financial Stability Board. (2024). The financial stability implications of tokenisation. https://www.fsb.org/2024/10/the-financial-stability-implications-of-tokenisation/

[22] See https://coinmarketcap.com/currencies/pax-gold/ and https://coinmarketcap.com/currencies/tether-gold/ for current market capitalization levels.

[23] Aquilina et al. (2023). Bank for International Settlements. Cryptocurrencies and decentralised finance: functions and financial stability implications. https://www.bis.org/publ/bppdf/bispap156.htm

insolvency. As the market continues to expand rapidly, the **absence of clear and harmonized global standards** poses challenges for both **investor protection and financial stability**.

2.2.2. Layer-2 infrastructure

The main blockchains (such as Bitcoin or Ethereum) are often referred to as **Layer-1 (L1)**, as they are the base protocol that maintains the core distributed ledger and consensus mechanism. L1 blockchains validate and record every transaction directly on-chain, ensuring security and decentralization. However, as network usage grows, these blockchains can become congested, leading to slower transaction times and higher fees.

To overcome the inherent limitations of L1 blockchains, such as congestion, high fees, and slower transaction times, developers have created **Layer-2 (L2)** solutions. L2 protocols operate on top of the base blockchain but handle most transaction processing off the main chain. By doing so, they can dramatically increase throughput and reduce costs while still anchoring security and consensus to the underlying L1 blockchain.

Layer-2 solutions take **different forms** depending on their design. Some, like rollups, aggregate or "bundle" multiple transactions together and post a single proof to the main chain, minimizing the amount of data and computation required on L1. Others, such as state channels or sidechains, allow users to conduct multiple interactions off-chain with only periodic settlement on L1. Across these designs, **L2 solutions maintain trustlessness and decentralization** while enabling faster, cheaper, and more scalable blockchain applications.

Layer-2

Why it matters:

Layer-2 solutions and tokenized assets are rapidly reshaping financial markets by **increasing transaction speed, lowering costs, and enabling broader participation** in DeFi.

However, they also create **regulatory blind spots**, **privacy-related challenges**, and **operational risks**.

While Layer-2 solutions facilitate **broader blockchain adoption**, they also introduce regulatory and supervisory **challenges**. Zk-rollups and other privacy-enhancing technologies obscure transaction-level data, **limiting visibility** for regulators, supervisors, and law enforcement, and **complicating compliance** with EU AML/CFT rules, the MiCA Regulation, and the FATF Travel Rule[24]. The growing use of Layer-2 networks for tokenized assets and stablecoins may create **regulatory blind spots**, heightening the need for adapted monitoring tools and risk assessment frameworks.

Since layer-2 networks rely on sequencers, validators, or operator sets that introduce potential **points of centralisation and failure, operational risks are emerging**. Disruptions or malicious activity at the L2 level could impact DeFi trading, lending, and collateralisation, with spillover effects on Layer-1 markets and financial stability. Interoperability between Layer-1 and Layer-2

[24] For further details, see Chainalysis https://www.chainalysis.com/blog/zero-knowledge-rollups-optimistic-rollups-overview/

networks, particularly through cross-chain bridges, remains a source of potential liquidity and settlement risk.

From a policy perspective, Layer-2 developments highlight the **tension between scalability and regulatory oversight**. Ensuring market integrity, traceability, and AML/CFT compliance in environments with reduced on-chain transparency will require innovative supervisory approaches and collaboration at EU and international levels. Layer-2 solutions are therefore a critical area for both technological innovation and regulatory adaptation.

2.2.3. Cross-chain bridges and multi-chain integration

While **Layer-2 solutions** focus on scaling single blockchains, **cross-chain bridges** address a different challenge: enabling the transfer of assets and liquidity **across distinct blockchain networks**. Despite this important distinction, similar to Layer-2 solutions, cross-chain bridges also represent potential systemic points of failure. Recently becoming a part of **critical infrastructure** in the multi-chain DeFi ecosystem, they also represent some of its **most acute vulnerabilities**.

Cross-chain bridges

Why it matters:

Cross-chain bridges are indispensable for the growth of DeFi and tokenized finance, yet their repeated failures make them **single points of systemic vulnerability**.

Ensuring their **resilience** is crucial for safeguarding **financial stability** in an increasingly interconnected blockchain landscape.

A series of major exploits, in which hundreds of millions of dollars were stolen, illustrates the scale of the problem: the Ronin attack[25] ($625 million), the Wormhole exploit[26] ($325 million), the Nomad hack ($190 million), the Harmony bridge breach[27] ($100 million), and the Qubit exploit[28] ($80 million) likely performed by North Korean hackers. Earlier, the Poly Network hack[29] of 2021, in which attackers stole over $610 million before returning most of the funds, remains a landmark case that exposed both technical weaknesses and governance gaps in decentralized infrastructures.

From a policy perspective, the frequency and magnitude of bridge exploits highlight **systemic risks** within multi-chain ecosystems. Because bridges connect otherwise isolated blockchains, a single failure can trigger **cascading effects across multiple networks**, amplifying contagion risks. This raises significant regulatory and supervisory questions around **operational resilience, liability**

[25] See BBC: https://www.bbc.com/news/technology-60933174

[26] See CBS News: https://www.cbsnews.com/news/wormhole-ether-cryptocurrency-320-million-hack/

[27] See TechCrunch: https://techcrunch.com/2022/06/24/harmony-blockchain-crypto-hack/

[28] See Chainalysis: https://www.chainalysis.com/blog/qubit-hack-north-korea/

[29] See Wikipedia: https://en.wikipedia.org/wiki/Poly_Network_exploit

allocation, and incident reporting in decentralized but high-value infrastructures, which are not yet fully covered by frameworks such as MiCA.

2.2.4. Decentralized governance

Building on the programmable logic of blockchains, **Decentralized Autonomous Organizations (DAOs)** represent one of the most significant governance innovations in the digital asset ecosystem. DAOs use smart contracts to manage collective decisions, including activities like protocol upgrades or liquidity incentives and infrastructure funding. The scale of assets under DAO governance has expanded rapidly: leading DAOs such as Uniswap, Aave, and Lido manage treasuries worth billions of euros, positioning them as **de facto financial institutions**.

Yet, despite their technological sophistication, DAOs remain subject to **deep legal and regulatory uncertainty**. U.S. courts have in recent rulings[30] **classified DAOs as general partnerships**, exposing token holders to unlimited joint liability[31]. At the same time, Wyoming's DAO Act (2021)[32] recognised DAO as legal entities, giving them some legal protection, but this does not apply elsewhere

In the EU, no dedicated legal framework exists, though aspects of DAO activity intersect indirectly with regulations such as MiCA[33] and the DORA[34] framework**,** neither of which directly addresses decentralized governance structures.

DAOs

Why it matters:

Decentralized Autonomous Organizations can bring significant innovation in decentralized governance, yet the **absence of a harmonized legal framework** and clear accountability creates uncertainty.

Addressing these gaps will be essential for **enabling innovation** in decentralized governance while **safeguarding financial stability**.

[30] See https://www.courtlistener.com/docket/65369411/commodity-futures-trading-commission-v-ooki-dao/ for the transcript of the relevant case, or a summary from CoinDesk: https://www.coindesk.com/policy/2023/06/09/cftc-wins-lawsuit-against-ooki-dao

[31] Additional info summarized here by Reuters: https://www.reuters.com/legal/government/column-blow-crypto-collectives-judge-rules-venture-backers-must-face-claims-2024-12-02/

[32] Full text of the Wyoming DAO bill: https://www.wyoleg.gov/Legislation/2021/SF0038

[33] Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets and amending Regulations (EU) No 1093/2010 and (EU) No 1095/2010 and Directives 2013/36/EU and (EU) 2019/1937. https://eur-lex.europa.eu/eli/reg/2023/1114/oj/eng

[34] Regulation (EU) 2022/2554 of the European Parliament and of the Council of 14 December 2022 on digital operational resilience for the financial sector and amending Regulations (EC) No 1060/2009, (EU) No 648/2012, (EU) No 600/2014, (EU) No 909/2014 and (EU) 2016/1011. https://eur-lex.europa.eu/eli/reg/2022/2554/oj/eng

2.3. Risks

Despite the already mentioned significant innovation efforts from the private sector as well as sizable global regulatory response, many of the risks already outlined above are still persistent and largely unsolved. Among others, these would constitute risks connected to centralization, integration with traditional finance, as well as market sentiment risks.

2.3.1. Centralization risk

Decentralization is a foundational principle of crypto and DeFi, yet many protocols and infrastructure layers exhibit notable **centralization vulnerabilities**.

Governance token ownership is often highly concentrated; for instance, the top 10 holders control roughly 53 percent of Uniswap governance tokens[35], with Compound[36] and Aave[37] showing similar patterns (37% and 53% respectively). DAOs also face concentration risks, as **voting power can be dominated by founders or early investors**.

Even more interestingly, on the Ethereum blockchain, two entities (BuilderNet and Titan Builder) produce 86 percent of all Ethereum blocks[38], representing a clear **monopoly and centralization point**. Critical infrastructure layers, including Layer 2 rollups, **sidechains**, and **block builders**, similarly rely on a limited number of sequencers or validators.

Cross-chain bridges further concentrate control, as a handful of key actors can influence multiple chains, amplifying systemic risk. Even **mining pools** and **major cloud providers**[39] hosting blockchain nodes can represent potential central points of failure.

Oracles, which connect off-chain data to smart contracts, demonstrate another concentration vector. Chainlink vastly dominates with approximately 84 percent market share within the Ethereum oracle market[40], while being integrated with over 450 projects spanning 21 blockchains[41]. Prior

Centralization

Why it matters:

Centralization across governance, infrastructure, and data oracles creates **single points of failure**, increasing the risk of manipulation, outages, or systemic disruption.

Understanding these centralization patterns is critical for designing **resilient DeFi systems** and effective **regulatory oversight**.

[35] See Etherscan for the top 10 holders of Uniswap: https://etherscan.io/token/tokenholderchart/0x1f9840a85d5af5bf1d1762f925bdaddc4201f984?range=10

[36] See Etherscan for the top 10 holders of Compound: https://etherscan.io/token/tokenholderchart/0xc00e94cb662c3520282e6f5717214004a7f26888?range=10

[37] See Etherscan for the top 10 holders of Aave: https://etherscan.io/token/tokenholderchart/0x7fc66500c84a76ad7e9c93437bfc5ac33e2ddae9?range=10

[38] See https://www.gate.com/learn/articles/monopoly-in-ethereum-block-builders-and-chain-abstraction-unveiling-profit-incentives-and-innovation-opportunities-in-the-blockchain-ecosystem/7690

[39] See Chainalysis: https://www.chainalysis.com/blog/crypto-hacking-stolen-funds-2024/

[40] See https://blockonomi.com/heres-why-chainlink-link-is-emerging-as-the-go-to-play-for-institutional-tokenization/

[41] See https://thecryptomagazines.com/chainlink-dominates-oracle-market-with-66-billion-in-secured-value/

incidents such as the bZx flash loan attack in 2020 show the potential consequences of **oracle manipulation**[42], highlighting the importance of this issue.

While the intent of these systems is decentralization, the practical landscape often favours a **small network of private actors** whose incentives focus on **profitability** rather than systemic stability or monetary policy objectives.

2.3.2. Links to traditional finance

The growing **overlap** between crypto markets and traditional finance (TradFi) introduces potential **systemic vulnerabilities**.

Institutional investment vehicles, including money market funds and asset managers, hold **substantial stablecoin positions**, with the USDC supply sitting around 72 billion. This reflects a 52 percent growth since the beginning of the year[43], largely dominated by institutional adoption[44]. **Tokenized real-world assets (RWAs)**, spanning U.S. Treasuries, corporate bonds, and commercial real estate, are increasingly represented on-chain, linking DeFi and TradFi[45].

Oracles, which supply on-chain protocols with off-chain data such as asset prices, interest rates, and FX rates, are a critical part of this integration. Already mentioned concentration risks in oracle networks mean that **disruptions, manipulation, or errors could propagate** and exacerbate shocks from DeFi directly to TradFi exposures[46].

Links to TradFi

Why it matters:

As crypto and TradFi become increasingly **entwined**, shocks in digital asset markets can **transmit** rapidly to regulated financial institutions, making **systemic oversight** and timely **intervention** essential.

Empirical evidence illustrates the **propagation of shocks from crypto to TradFi**. The Terra/Luna token collapse in 2022 led to liquidity strains across crypto hedge funds and stressed regulated investment funds with indirect exposures[47]. Similarly, stablecoin de-pegging events can disrupt

[42] For further details on the attack see https://www.coindesk.com/tech/2020/02/27/the-defi-flash-loan-attack-that-changed-everything

[43] See https://usdc.cool/ for the current supply of USDC.

[44] See CoinMarketCap: https://coinmarketcap.com/academy/article/usdc-nears-dollar61-billion-supply-in-2025-as-institutional-demand-outpaces-usdt

[45] See Redstone Finance: https://blog.redstone.finance/2025/08/07/redstone-h1-2025-recap-leading-oracle-innovation-in-defi-rwas-and-tokenization/

[46] See ChainLink for further details: https://chain.link/education-hub/market-manipulation-vs-oracle-exploits

[47] Cornelli G. et al (2023). Bank for International Settlements. Crypto shocks and retail losses. https://www.bis.org/publ/bisbull69.htm

payment systems, short-term funding markets, and liquidity management within banks holding these assets[48].

Without **mandatory reporting** of crypto holdings by traditional financial actors, regulators cannot precisely quantify the effects on individual balance sheets, only a general aggregated picture.

2.3.3. Volatility risk

Crypto markets are **highly volatile**, reflecting thin liquidity in many tokens and the absence of safeguards typical in regulated markets, such as circuit breakers, short-selling restrictions, or centralized clearing mechanisms[49]. **Price swings** can be amplified by algorithmic trading, DeFi leverage, and sudden liquidity withdrawals. For example, the Luna/Terra collapse in May 2022 wiped out over $40 billion in market value within days, **cascading** through multiple exchanges and DeFi protocols[50]. Similarly, the FTX collapse in November 2022 triggered sharp declines and liquidity strains across leveraged stablecoin and tokenized asset positions[51].

Sentiment-driven dynamics also strongly influence crypto prices. Social media platforms, forums, and community channels, including **Reddit**, **Discord**, and **Twitter**, play a central role in **coordinating investor behavior**. Memecoins such as Dogecoin and Shiba Inu have experienced 30–50% intraday price swings following viral activity or endorsements by influencers, such as Elon Musk[52]. During the 2020–2021 DeFi Summer, **coordinated social media campaigns** and **community-driven hype** around yield farming led to extreme token price volatility[53] and rapid capital flows into new protocols[54].

Volatility risk

Why it matters:

Volatile and in many cases sentiment-driven crypto **price swings** have the potential to transmit to traditional financial institutions, potentially disrupting liquidity, funding conditions, and the valuation of assets across the broader financial system.

Without real-time **monitoring** and **risk mitigation**, these volatile channels remain largely unchecked

[48] Bains, P. et al. (2022). International Monetary Fund Fintech Notes. Regulating the Crypto Ecosystem: The Case of Stablecoins and Arrangements. https://www.imf.org/en/Publications/fintech-notes/Issues/2022/09/26/Regulating-the-Crypto-Ecosystem-The-Case-of-Stablecoins-and-Arrangements-523724

[49] For a more detailed analysis on missing levers and checks in crypto markets, see https://northpennnow.com/news/2025/may/23/why-is-crypto-so-volatile-a-day-traders-guide-to-market-fluctuations/

[50] Cornelli G. et al (2023). Bank for International Settlements. Crypto shocks and retail losses. https://www.bis.org/publ/bisbull69.htm

[51] See The Financial Times: https://www.ft.com/content/99841273-4045-49ff-90a2-d17df28fcfc9

[52] An article detailing the price surge in dogecoin after endorsement by Musk: https://www.fortuneindia.com/investing/dogecoin-surges-30-after-elon-musk-changes-twitter-logo-to-shiba-inu/112127

[53] Predicting these market movements would require monitoring of social networks, community sentiment, and viral trends.

[54] See Messari: https://messari.io/report/q2-21-defi-review

In addition, there are emergent and potentially disruptive technologies which also pose risk to the price stability within crypto markets. **Artificial intelligence (AI)** presents complex risk vectors and novel operational dynamics. For instance, in early 2025, the cryptocurrency market experienced a significant flash crash, erasing over $269 billion in market capitalization[55]. AI and meme tokens were among the hardest hit, with the crash attributed to the launch of the Chinese AI app DeepSeek.

Moreover, **Generative AI models** might soon facilitate the creation of synthetic identities that can bypass conventional AML controls, thereby enabling fraud, money laundering, and identity theft[56] on crypto platforms[57]. Additionally, autonomous AI agents capable of **interacting with DeFi protocols** autonomously raise the prospect of sophisticated market manipulation and governance attacks conducted without direct human intervention.

These **emergent risks** necessitate the development of advanced analytical methodologies, adaptive regulatory frameworks, and enhanced international collaboration to effectively monitor, assess, and mitigate AI-crypto intersection risks[58].

2.3.4. Monetary sovereignty risk

The **predominance of the US dollar** in crypto markets is underpinned not only by established network externalities but also by legislative frameworks. For example, the U.S. Digital Asset Market Structure and Investor Protection Act[59]provides regulatory clarity and incentivizes innovation in stablecoins, exchanges, and DeFi platforms. More recently the GENIUS Act[60] aims to **enhance the dollar's currency status by driving demand for U.S. Treasuries via dollar-denominated stablecoins**.

Conversely, the **European Union has yet to establish** a comparably developed **euro-denominated stablecoin** ecosystem. In parallel, the European Central

Monetary sovereignty risk

Why it matters:

The **EU reliance on dollar-denominated stablecoins** limits its control over monetary policy and exposes European markets to U.S. regulatory and geopolitical influence, potentially **undermining both financial stability and monetary sovereignty**.

Accelerating the development of **euro-based digital assets** and fostering a robust **euro stablecoin ecosystem** are essential for enhancing the EU's strategic autonomy and regulatory influence in global crypto finance.

[55] See https://www.ccn.com/news/crypto/crypto-flash-crash-850m-market-chinese-deepseek-blame/

[56] Crypto platforms are not the only platforms potentially susceptible to these risks, but they do tend to allow less recourse to the victim.

[57] See https://www.fatf-gafi.org/en/publications/Fatfrecommendations/targeted-update-virtual-assets-vasps-2024.html

[58] G'Sell, F. (2024). Stanford Cyber Policy Center. Regulating Under Uncertainty: Governance Options For Generative AI. https://cyber.fsi.stanford.edu/content/regulating-under-uncertainty-governance-options-generative-ai

[59] See https://www.congress.gov/bill/118th-congress/house-bill/5745

[60] See https://www.congress.gov/bill/119th-congress/senate-bill/1582/text

Bank's Digital Euro initiative, while progressing, remains in early pilot stages with broader adoption not anticipated before 2029[61].

This regulatory and infrastructural asymmetry exposes the EU to marginalization in global crypto finance governance and diminishes its **capacity to influence emerging standards** and innovation trajectories in tokenized finance. Furthermore, the EU's dependence on dollar-denominated crypto assets **increases exposure to extraterritorial regulatory and geopolitical pressures** originating from the United States. Without a **strategic response**, European **monetary sovereignty** and **financial stability** could gradually erode[62] as a result.

[61] European Central Bank. (2025). https://www.ecb.europa.eu/euro/digital_euro/html/index.en.html

[62] Schaaf J. (2025). European Central Bank. From hype to hazard: what stablecoins mean for Europe. https://www.ecb.europa.eu/press/blog/date/2025/html/ecb.blog20250728~e6cb3cf8b5.de.html?s=09

3. The CIPHER framework

Against the backdrop summarized in the previous chapter, the **CIPHER framework** intends to provide a systematic approach to **anticipate, monitor and analyse** vulnerabilities and tension points across crypto and DeFi markets. Its strength lies in enabling **early identification of emerging risks**, supporting evidence-based **policy advice**, and informing the Commission's cross-DG initiatives **on financial stability, digital finance and innovation**. By focusing itself around blockchain data and analytics, CIPHER intends to equip analysts and policy officers alike with a granular, forward-looking view of the crypto ecosystem.

3.1. Architecture

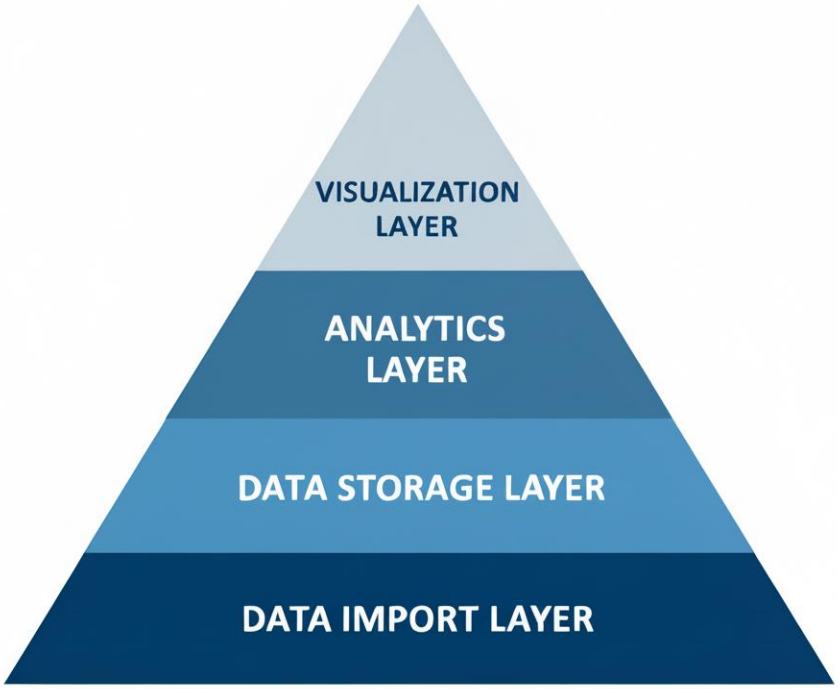
The CIPHER framework consists of four distinct layers - **Data Import, Data Storage, Analytics and Visualization (Figure 1)**.

The first two - **Data Import** and **Data Storage** - are the core of the framework, allowing for live data ingestion directly from the Bitcoin and Ethereum blockchain and its storage in a dedicated data warehouse.

The third layer - **Analytics** - allows user access to the data and the ability to run SQL queries on it. This translates to a high level of freedom in terms of data analysis and allows the possibility of conducting complicated operations, including creation of new tables and / or views.

The final layer is **Visualization**, which provides multiple charts and statistics about the crypto landscape that have been sourced from the underlying data. This layer is represented by a user-friendly graphical interface which is easily accessible via a standard internet browser, with no prior knowledge in data analysis of programming required.

Figure 1. Graphical illustration of the four layers of the CIPHER framework



Source: own graphic

3.1.1. Data import layer

From the **data import** perspective, the CIPHER infrastructure currently employs **two dedicated full nodes**, one for **Ethereum** (ETH) and one for **Bitcoin** (BTC), providing direct, trustless and permissionless access to raw blockchain data. The real-time delay of the database with respect to the current state of the blockchain is approximately one hour for Bitcoin and 20 minutes for Ethereum.[63]

In addition, CIPHER currently integrates **off-chain data** by retrieving historical **exchange rates** between fiat currencies and stablecoins through the Binance API. This provides essential valuation context for stablecoins and cross-asset relationships. For enhanced **entity resolution**, the system uses proprietary APIs from private blockchain analytics companies to **attribute blockchain addresses** to known actors (e.g. exchanges).

Data ingestion and transformation workflows are managed by a dedicated development server, where custom scripts **extract** and **normalize** data from nodes and APIs before forwarding it to the **storage layer**.

3.1.2. Data storage layer

Once ingested and transformed, the development server forwards the data into a centralized database server for **data storage**. This database is optimized for high-performance analytical queries and multi-dimensional time-series data, allowing users to work with **terabytes of data** and **billions of transactions**. It currently houses comprehensive BTC and ETH blockchain data, including block-level transactions for the Bitcoin blockchain or event-level information from the Ethereum blockchain.

The database schema is designed to support **complex queries** across different temporal scales and data types, laying the foundation for a **wide range of crypto analytics**. Currently, each blockchain is represented by its own schema with specific tables representing the transaction and/or event data, reflecting the relevant blockchains' specificities.

For a more technical overview of the database structure, see **Annex 2**.

3.1.3. Analytics layer

The **analytics layer** of the CIPHER framework is focused on providing a comprehensive solution for data scientists and data analysts who prefer to run queries directly on the granular data. As such, it allows remote connection to the database and **direct querying** of the underlying data via standard SQL syntax.

[63] Bitcoin blocks are ingested roughly every 10 minutes and Ethereum data every 32 blocks (~6 minutes). To ensure reliability, CIPHER does not ingest blocks immediately: for Bitcoin, it waits for six confirmations, and for Ethereum, until reorgs would require over one-third of validators to lose their stake. This aligns with industry standards for mitigating chain reorganizations ("reorgs") and ensures that ingested data is both stable and highly reliable for downstream analysis.

For this purpose, the database schema already includes several **pre-computed tables** that have been prepared and indexed to make the user access easier and more streamlined.

This layer also includes pre-computed tables connected to **DeFi event tracking,** currently focused on key Ethereum protocols such as Uniswap and Compound. Smart contract logs corresponding to these protocols are parsed to monitor minting of tokens, their redemption, governance proposals, votes, upgrades, and significant market events. This continuous parsing and pre-computation enables **real-time situational awareness** about the relevant protocols based entirely on **on-chain data**.

3.1.4. Visualization layer

The **visualization layer** leverages the CIPHER framework's ingested and stored data to generate a suite of indicators and insights. These include for instance **monitoring reserve holdings** of major centralized exchanges, or **stablecoin capitalization and peg status**.

Visualization and user interaction are provided through **a web browser dashboards**, offering an intuitive interface for **real-time monitoring** with no prior knowledge of programming required. Dashboards support multi-scale zooming, allowing users to inspect data over multi-year horizons down to individual block details. **Filtering** functionalities enable **efficient querying** by protocol, time frame, or entity and additional options and visualization metrics are expected to be added soon.

For a more detailed overview as well as screenshots of the visualization layer, see **Annex 3**.

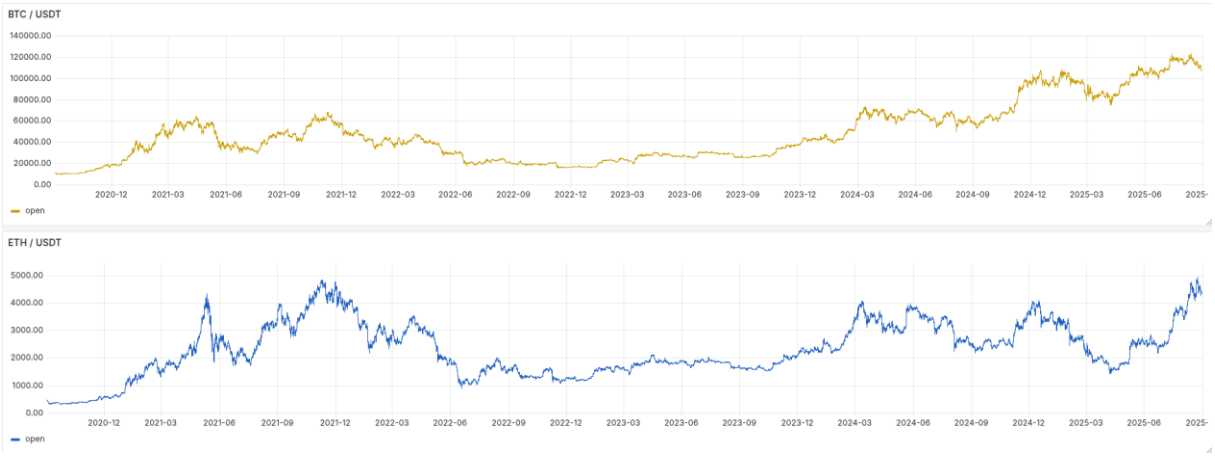
3.2. Early use cases and insights

The development of CIPHER has already enabled the visualization of several indicators that demonstrate both the system's analytical capabilities and its relevance for monitoring the crypto markets. These early use cases provide insight into market concentration of centralized exchanges and the evolving role of stablecoins, while highlighting the potential for policy-relevant applications.

3.2.1. Crypto token price data

Crypto **token price** data measured in relation to fiat are a core input for monitoring crypto markets and their links to traditional finance. **Figure 2** shows an example visualization from the CIPHER dashboard, where the BTC/USDT and ETH/USDT pairs highlight the central role of dollar-denominated stablecoins as the settlement layer of the crypto ecosystem. Tracking these pairs within CIPHER provides insight not only into market valuation but also into the structural dependency of crypto markets on U.S. dollar liquidity.

Figure 2. Price of BTC and ETH in relation to USDT during the previous five years.



Source: Binance market data, CIPHER visualization dashboard

From a policy perspective, monitoring price dynamics helps assess how shocks in crypto may spill over to traditional finance. Sharp swings in BTC or ETH can trigger liquidations in DeFi protocols and shift demand for stablecoins, which in turn may affect traditional financial instruments used as backing. For this reason, CIPHER treats robust price data as an essential foundation for evaluating transmission channels between crypto and traditional finance.

3.2.2. Balances of centralized exchanges (CEXs)

CIPHER monitors **balances** of a hand-picked set[64] of **centralized crypto exchanges**, which helps with the understanding of the liquidity conditions within the crypto ecosystem. For an example of such balance monitoring, refer to **Figure 3** for an overview of Binance USDT holdings. This figure is taken directly from the CIPHER's visualization dashboard and shows a significant increase in Binance's USDT holdings between November and December 2024 because of heightened institutional interest, increased market liquidity, and strategic positioning.

Centralized exchanges continue to act as the main **gateways** between the crypto market and the traditional financial system, holding large amounts of customer deposits in stablecoins and other liquid assets. Monitoring the balances of major venues therefore allows an assessment of where liquidity is concentrated, how it is distributed across different market actors, and how these patterns evolve over time. Particular attention has been given to reserves in USDT and USDC, as these two stablecoins function as the principal settlement assets in most trading pairs.[65]

This functionality strengthens CIPHER's ability to track exchange-level risk exposures. For example, sudden outflows of stablecoins from a single exchange may signal a loss of confidence, liquidity

[64] Exchanges Binance, Bybit, OKX, and Coinbase are included at this moment with holdings of USDC and USDT. These already represent the majority of the market: https://www.cryptotimes.io/2025/07/16/binance-okx-bybit-bitget-led-crypto-markets-in-q2-2025/

[65] The system is designed to be easily extendable to additional exchanges and tokens, supporting broader coverage of global markets as policy needs evolve. However, this functionality requires continuous operational maintenance, as exchanges regularly change the blockchain addresses used for custody. Keeping attribution accurate and up to date is therefore a critical technical task, underlining the need for robust infrastructure and sustained resources to ensure reliable monitoring.

stress, or the reaction to regulatory developments. Conversely, large inflows may point to increasing market concentration, raising questions about resilience in the event of an operational outage, cyber incident, or policy intervention. By systematically collecting and visualising these balances, CIPHER provides a vantage point on vulnerabilities that could propagate to European market participants through reliance on a limited number of non-EU venues.

Figure 3. Overall holdings of USDT by the Binance exchange.



Source: Ethereum blockchain data, CIPHER visualization dashboard

3.2.3. Stablecoins

Just like centralized exchanges, **stablecoins** also constitute a core element of CIPHER's monitoring, as they are the principal **bridge** between crypto-assets and traditional financial markets. By tracking volumes in major stablecoins[66], CIPHER provides insights into which settlement assets dominate the market and how their usage evolves over time.

Monitoring stablecoins is particularly relevant because their adoption reflects both market sentiment and regulatory conditions. For instance, **Figure 4** shows a recent example from the CIPHER visualization dashboard – a **sharp increase in the USDe stablecoin's market capitalization** following the passage of the US' **GENIUS Act** on July 18th, 2025.

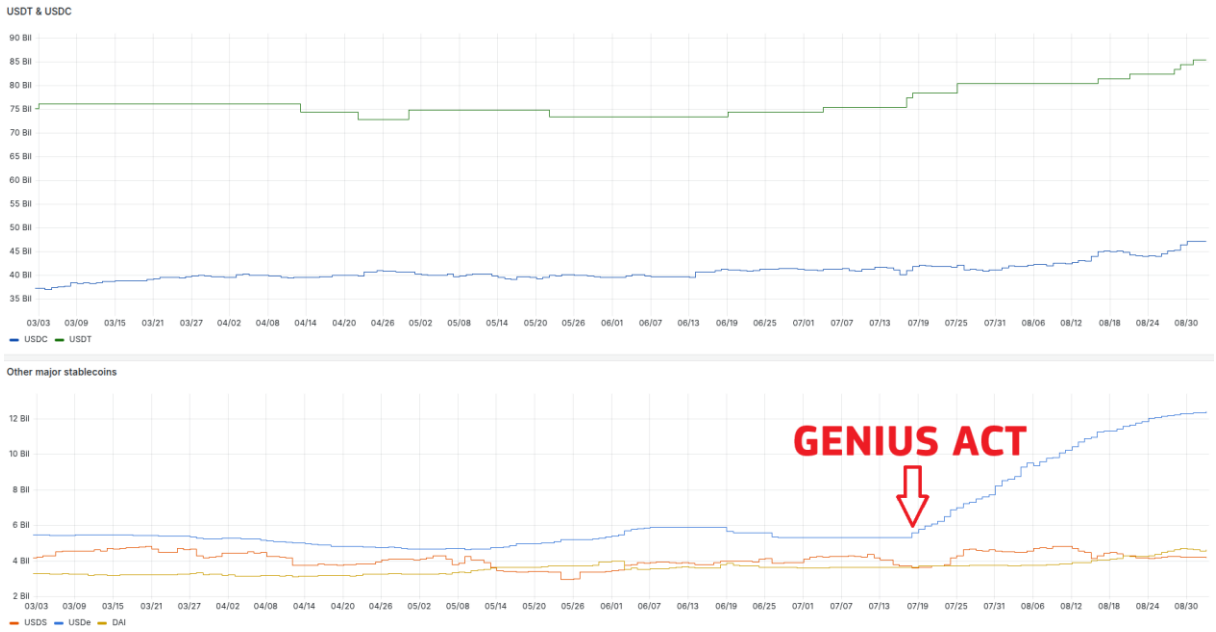
The measure was widely interpreted as supportive of the issuer's business model, and the market capitalization nearly doubled in the immediate aftermath. This extraordinary growth was driven by the fact that the GENIUS Act prohibits stablecoin issuers from offering interest directly to holders, creating demand for alternatives that provide yield through decentralized mechanisms instead[67]. USDe fits this niche, offering staking and yields through DeFi platforms, tapping unmet investor demand despite the tightened regulatory environment.

This episode illustrates how regulatory interventions can have an immediate and measurable impact on crypto market dynamics. By integrating such policy signals with on-chain data, CIPHER makes it possible to study the direct links between regulation, adoption, and systemic risk.

[66] At the moment, the framework is tracking USDC, USDT, USDS, USDe, and DAI stablecoins. The infrastructure is designed to be readily extendable to additional tokens, ensuring adaptability as the market landscape shifts.

[67] See this article for further details: https://cointelegraph.com/news/yield-bearing-stablecoin-supply-surges-after-genius-act

Figure 4. Volume of USDT, USDC, USDS, USDe and DAI stablecoins. Note the doubling in USDe volume following directly the "GENIUS" Act from July 18th, 2025 (lower figure, blue line).



Source: Ethereum blockchain data, CIPHER visualization dashboard

Stablecoin data also provides important signals for financial stability monitoring. Because stablecoins are typically dollar-denominated, their widespread use in European markets deepens the reliance on non-euro settlement assets, with potential implications for monetary sovereignty. Moreover, shifts in stablecoin volumes can foreshadow liquidity stresses, contagion risks, or the build-up of leverage within DeFi protocols. The ability to observe these developments in near real-time through CIPHER therefore represents a key step toward building an evidence base for informed policy decisions.

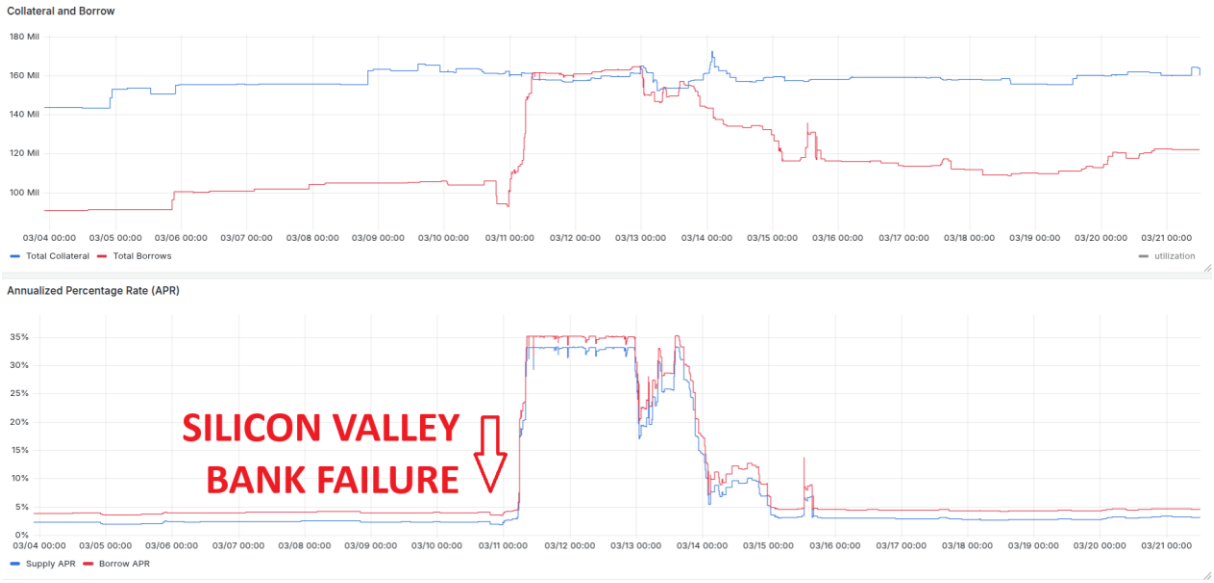
3.2.4. DeFi protocols

The CIPHER framework has been designed to **monitor activity across DeFi protocols** in real time. In principle, **any protocol deployed on Ethereum can be integrated** into the framework, as all relevant data are publicly available on-chain. At present, monitoring modules have been implemented for **Uniswap** (decentralized exchange) and **Compound** (lending and borrowing). These provide an operational proof of concept for extracting, processing, and visualizing protocol-level data streams.

DeFi lending protocols such as Compound operate on the basis of **algorithmic interest rate** models. Borrow and supply rates adjust **automatically** in response to utilization: when borrowing demand for a token increases relative to available supply, interest rates rise, and vice versa. This makes such platforms a sensitive indicator of liquidity conditions and stress in the broader crypto ecosystem.

As an example, **Figure 5** taken from the CIPHER monitoring dashboard, shows the amount of collateral available and the resulting interest rates for the Compound protocol in March 2023. During this time, the **utilization rate of USDT rose sharply**, which caused borrowing rates to spike to unusually high levels, while the protocol was virtually drained of its USDT liquidity.

Figure 5. Increase in USDT borrowing and the associated spike in borrowing and lending rates of the USDT stablecoin as a result of the Silicon Valley Bank failure.



Source: Ethereum blockchain data, CIPHER visualization dashboard

The reason behind this event was the collapse of the **Silicon Valley Bank (SVB),** which led to a temporary loss of confidence in the USD Coin (USDC) stablecoin. At the time, Circle, the issuer of USDC, held a portion of its reserves with SVB. Following the bank's failure, USDC traded down to around $0.88 on secondary markets.

This de-peg created a profitable loop:

1. Market participants borrowed **USDT** (Tether) from Compound.
2. They swapped the borrowed USDT into discounted **USDC** on decentralized exchanges.
3. The USDC was then **redeemed with Circle** at its face value of $1, realizing an immediate profit.
4. The redeemed dollars were reintroduced into crypto markets, allowing traders to obtain more USDT and repeat the cycle.

Each round of this strategy **was highly profitable**, and traders executed it at scale. Importantly, the ability to redeem USDC for US dollars is still limited to large institutional clients with compliant KYC regulation, making this operation practically inaccessible to retail investors.

This episode illustrates that shocks originating in the traditional financial system (e.g. SVB's failure) can **transmit** rapidly into DeFi markets and that automated interest rate models in lending protocols are directly exposed to such flows, providing measurable **signals** of stress and arbitrage. CIPHER can capture these dynamics as they unfold, highlighting **cross-market linkages** between banking, stablecoins, and DeFi.

Together, these insights demonstrate the value of extending systematic monitoring to a broader set of protocols. While the current focus has been on Uniswap and Compound, the same approach will be scaled to cover other major DeFi applications (e.g. Aave, MakerDAO), thereby building a comprehensive view of risk transmission channels within the crypto-financial ecosystem.

3.3. Potential applications for policy and regulation

This section provides further details on integrating CIPHER into policy workflows. Some of these functionalities are already being developed, while others are subject to agreement with the interested stakeholders.

3.3.1. Integration into policy workflows

The CIPHER framework is being developed to integrate directly into policy workflows by providing automated monitoring, analytics, and reporting that transform raw blockchain and market data into **policy-relevant evidence**. The aim is to embed crypto-asset monitoring into the broader policy cycle, from early **evidence gathering** to **supervisory oversight** and **ex-post evaluation**.

A key feature is the ability to deliver insights at different levels of granularity. **High-level dashboards** can provide policymakers with intuitive overviews of market developments via CIPHER's visualization layer, while technical teams can access more detailed data for **customized analysis** via the analytics layer. This **dual-access model** ensures that the same infrastructure can support both **strategic briefings** and detailed **analytical work**.

Examples of outputs that have already been developed or that are to be developed soon include:

- **Stablecoin monitoring dashboards** focused on peg stability and deviations.
- **DeFi risk dashboards** highlighting liquidity trends, collateralisation levels, and protocol-specific insights.
- **Exchange balance monitoring** to assess the solvency and liquidity position of major centralized exchanges, enabling early-warning signals of stress.
- **Macro-level indicators** of market capitalisation, transaction costs, and liquidity, which can be correlated with traditional financial assets to assess contagion pathways.

Beyond monitoring, CIPHER can also provide **automated reporting capabilities**. For example, when a stablecoin experiences stress or when DeFi borrowing surges unexpectedly, pre-configured alerts could generate concise summaries. This would reduce the reliance on manual data gathering

Integration into policy workflows

Why it matters:

Combining **market monitoring** with **automated reporting**, and granular **data analysis**, CIPHER has the potential to become a structured **evidence base** for policymaking on crypto and decentralized finance.

This will allow the EU policy institutions to move from ad hoc assessments toward continuous, **data-driven monitoring** and strengthen the capacity to **anticipate risks** rather than merely respond to them.

and allow for faster institutional response. In this spirit, there are multiple near-term enhancements expected soon which will expand the system's applicability for policy users[68].

3.3.2. Financial stability monitoring and stress testing

A specific field in which CIPHER's capabilities can be leveraged is financial stability monitoring of the crypto and DeFi landscape. By capturing dimensions such as liquidity, leverage, and interconnections within the crypto-asset ecosystem, the system has the potential to generate early signals of market vulnerabilities, including those that may transmit to the wider financial system.

The monitoring component is designed to cover key systemic risk channels, including:

- **Stablecoin stability:** Monitoring for de-pegging events that may undermine settlement reliability and confidence in crypto-based payment mechanisms.
- **Decentralized lending risks:** Identifying shocks in lending protocols, such as large-scale liquidations or collateral shortfalls, which can trigger procyclical feedback loops across DeFi markets.
- **Cross-market liquidity flows:** Analysing stress propagation across exchanges, where disruptions in one venue can rapidly spill over to the broader ecosystem.

DeFi stress testing

Why it matters:

CIPHER can provide policymakers with **early warning signals** of **vulnerabilities** in crypto and DeFi markets by using blockchain-native data and smart contract rules.

Unlike traditional stress tests, it enables **high-fidelity simulation** of shocks directly within the blockchain environment, offering a transparent **tool** to assess **contagion** risks and **systemic** resilience.

In addition to this ongoing monitoring, the team behind CIPHER is actively exploring the development of a **dedicated stress testing framework** to support forward-looking analysis of DeFi markets. The aim is to create a **simulation environment** where hypothetical shocks can be introduced directly at the blockchain node level, thereby ensuring that the scenarios reflect the actual logic of smart contracts and the operational mechanics of decentralized protocols. Such a framework will allow the injection of **stress scenarios** directly into the execution environment of DeFi protocols, for example by simulating abrupt price shocks, sudden collateral withdrawals,

[68] For instance, a currency selector will enable multi-denomination analysis, allowing indicators to be expressed in euro, U.S. dollar, or ETH depending on the policy context.

Box 1: Disruption scenario for a stress test: a stablecoin shock

Disruption scenario for a stress test: a stablecoin shock

A potential disruption scenario to stress-test the DeFi landscape would be a **shock to a stablecoin price**. As of mid-2025, the two dominant stablecoins USDC and USDT represent almost $250 billion in circulating supply[69].

The failure of TerraUSD in May 2022 exemplifies the systemic risks inherent in stablecoin designs. The loss of its peg precipitated a $40 billion market correction, widespread liquidations, and a marked **erosion of confidence** in algorithmic stablecoins[70].

This event triggered rapid shockwaves across interconnected DeFi lending protocols. Likewise, centralized exchanges holding stablecoins[71] as reserve assets experienced solvency strains, thereby amplifying systemic market stress.

Furthermore, the increasing integration of stablecoins into traditional financial instruments, including tokenized assets and ETFs, raises the potential for transmission of crypto market shocks into regulated financial markets[72].

validator concentration, or network-level liquidity fragmentation (see **Box 1** [69,70,71,72] and **Box 2**[73,74] for examples).

Running these scenarios directly on the node ensures that the stress test captures the full chain of protocol responses, including liquidations, collateral auctions, protocol-level stabilisation mechanisms such as fee structures or rate adjustments, or liquidity reallocation across venues. This approach provides a high-fidelity view of how shocks **propagate through smart contracts and across interconnected protocols**, reducing reliance on simplified assumptions or aggregate approximations.

This approach differs fundamentally from traditional financial stress testing. In conventional settings, stress tests rely on supervisory reporting, balance sheet data, and model-based approximations of institutions' behaviour under adverse conditions. To satisfy that, heuristic

[69] See for instance https://stablecoin.com/market-cap/

[70] Briola, A. et al. (2023). Anatomy of a Stablecoin's failure: The Terra-Luna case. *Finance Research Letters.* https://www.sciencedirect.com/science/article/pii/S1544612322005359

[71] See the financial times: https://www.ft.com/content/5a3b6fe9-329c-4c97-9f2b-3b00edeb0618

[72] Agur, I. at al. (2025). International Monetary Fund. Tokenization and Financial Market Inefficiencies. https://www.imf.org/en/Publications/fintech-notes/Issues/2025/01/29/Tokenization-and-Financial-Market-Inefficiencies-561256

[73] See Etherscan for the top 10 holders of Uniswap: https://etherscan.io/token/tokenholderchart/0x1f9840a85d5af5bf1d1762f925bdaddc4201f984?range=10

[74] See Wikipedia: https://en.wikipedia.org/wiki/Poly_Network_exploit

Box 3: Disruption scenario for a stress test: a protocol failure

Disruption scenario for a stress test: a protocol failure

Another potential scenario for an origin of DeFi stress test would be a major **protocol failure**.

Decentralized Autonomous Organizations (DAOs) and DeFi protocols administer substantial volumes of user funds via smart contracts and on-chain governance mechanisms. However, **governance power concentration** introduces significant vulnerabilities. For instance, Uniswap governance tokens are disproportionately controlled by a limited cohort representing over 50% of voting power[73].

The 2021 Poly Network exploit, involving a $600 million loss through a cross-chain bridge vulnerability, underscores the technical risks embedded within complex smart contract architectures[74].

constants are usually inserted by policy makers to judge the extent of asset value depreciations in fire sale scenarios, or to judge the extent of recovery.

By contrast, in decentralized finance **all relevant data are natively recorded on the blockchain,** and the **behavioural rules of protocols are codified in smart contracts,** making the system much more deterministic. This allows the stress testing framework to simulate scenarios directly within the blockchain execution environment, without requiring detailed assumptions about how participants or intermediaries might respond in crisis. The node can **replay shocks** against the actual transaction history and protocol logic, ensuring that the propagation of stress across markets reflects the precise rules governing liquidations, collateral auctions, interest rate adjustments, and liquidity flows. This results in a **high-fidelity**, fully **transparent** stress testing methodology that is unique to blockchain-based financial markets.

This stress testing framework has the potential to serve as a **policy-relevant laboratory** for scenario analysis. It will enable quantification of vulnerabilities and assessment of resilience under a range of adverse conditions, offering insights into:

- The speed and scale of contagion dynamics across protocols and exchanges.
- Threshold effects and explorations into scenarios where small shocks might trigger disproportionate systemic reactions.
- The role of protocol design (e.g. collateral requirements, liquidation rules, validator incentives) in amplifying or dampening stress.

By providing the ability to "replay" shocks under **controlled conditions**, the framework would allow policymakers to evaluate the robustness of decentralized market infrastructures and to compare different stress scenarios on a consistent basis.

3.3.3. Strategic autonomy and monetary sovereignty

In addition to financial stability questions, CIPHER can also support the assessment of Europe's **strategic autonomy in digital finance** by systematically monitoring dependencies on non-EU stablecoins, exchanges, and DeFi protocols.

The widespread use of foreign-issued assets and platforms introduces potential vulnerabilities, as regulatory interventions, market disruptions, or geopolitical tensions outside the EU could propagate directly to European market participants. Key transmission channels include payment and settlement exposure, where **reliance on non-EU stablecoins** may affect the stability and efficiency of euro-denominated transactions, as disruptions to foreign-pegged assets could impede settlement finality or trigger confidence losses. **Liquidity concentration in non-EU exchanges or protocols** can amplify volatility and fragmentation, as stress or outages in one venue may rapidly propagate to European participants[75]. **Collateral and leverage risks** arise when foreign-issued assets are used as collateral in lending or derivatives protocols, exposing EU users to valuation shocks or automatic liquidations triggered by collateral shortfalls in protocols operating outside EU jurisdiction. Moreover, if **USD-denominated stablecoins are the primary bridge** between DeFi and the traditional financial system, shocks affecting these assets could indirectly influence euro-area participants, highlighting a structural **channel of exposure** that may be relevant for assessing financial and monetary policy risks.

Strategic autonomy

Why it matters:

CIPHER intends to monitor the **adoption of euro-denominated stablecoins** and the DeFi markets' reliance on USD-pegged assets, helping policymakers assess risks to Europe's **monetary sovereignty** and the potential for external shocks to affect the EU's digital financial ecosystem.

Near-term enhancements to CIPHER aim to strengthen the monitoring of these channels by tracking the **adoption of euro-denominated stablecoins** and synthetic assets, which allows for an evaluation of whether EU-issued alternatives are gaining traction relative to foreign benchmarks. By integrating these metrics into a structured monitoring framework, CIPHER intends to provide policymakers with actionable insights into dependencies, potential transmission channels, and live evaluation of the **robustness** of the EU's digital finance ecosystem.

[75] While the MiCA Regulation establishes comprehensive requirements for EU-based crypto-asset issuers, service providers, and stablecoins, it does not directly address systemic exposures stemming from non-EU exchanges or protocols. Dependencies on foreign platforms and USD-denominated stablecoins may therefore still create transmission channels of volatility and liquidity risk that fall outside the regulatory perimeter of MiCA.

3.3.4. Unfair practices

DeFi and crypto markets exhibit a range of **unfair practices** that can undermine market integrity and investor confidence, many of which have analogues in traditional finance but remain **largely unregulated** in this space.

Among these, **Maximal Extractable Value (MEV)**[76] stands out as a particularly pervasive topic, despite being a quintessential part of the blockchain ecosystem. MEV naturally arises when actors optimize the transaction ordering within a new block to maximize the value that they receive from such operation. In some cases, such reorders, insertions, or exclusions of transactions are compared to unfair practices that have been long banned in traditional finance. Among those, the most well-known are **front-running** or **sandwich attacks**, in which the builder strategically places their own transactions just before / after a large transaction of the victim, effectively exploiting the effect such large transaction might have on the market[77].

Unfair practices

Why it matters:

Front-running, wash trading, and similar practices distort prices and concentrate profits among a few select actors, harming ordinary users and undermining market integrity.

CIPHER intends to **quantify** these behaviors to inform a potential policy or regulatory response.

CIPHER's access to complete blockchain data allows for the analysis of transaction ordering, gas fees, and sequences of related transactions, enabling the **identification** of these practices. In addition, a consumer-centric view could be applied to estimate such cost of MEV strategies to the "victim".

A specific form of MEV extraction called **Front-running** has been banned in traditional financial markets for decades. It spans any on-chain environment where transaction ordering matters, such as automated market maker (AMM) protocols or lending platforms, where actors exploit knowledge of pending transactions to gain advantage, often harming other users. Using timestamps, transaction ordering, and repeated behavioral patterns, CIPHER can attempt to better understand vulnerabilities of these protocols, which systematically profit from such positioning.

Closely related are **wash trading** and **pump-and-dump** schemes, which **manipulate** market activity and asset prices to create false signals or generate profit at the expense of unsuspecting participants. Wash trading can be detected through the analysis of circular or repetitive interactions between wallets, while pump-and-dump behavior can be inferred by linking clusters of activity to rapid price movements. By quantifying these patterns, CIPHER can highlight periods of abnormal market activity offering regulators a high-level view of manipulative behaviours.

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[76] Barczentewicz, M. (2023). MEV on Ethereum: A Policy Analysis. *ICLE White Paper.* https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4332703

[77] European Securities and Markets Authority. (2023). Consultation Paper on technical standards and guidelines specifying certain requirements of the Markets in Crypto Assets Regulation (MiCA) on detection and prevention of market abuse, investor protection and operational resilience.

Similarly, **liquidity and oracle manipulation** can create systemic risks, as actors influence on-chain prices or collateral data to trigger liquidations or arbitrage opportunities. CIPHER can track abrupt changes in liquidity pools, collateral ratios, and price feeds, linking these anomalies to specific wallet clusters to assess exposure to manipulation at the market-wide level.

Finally, even ostensibly benign protocol features, such as **fee rebates** or **staking incentives**, can be optimized by sophisticated actors to maximize gains. CIPHER can monitor transaction frequency, wallet activity, and net benefits from such mechanisms, quantifying their extent and concentration.

Taken together, these capabilities will allow CIPHER to provide a **comprehensive, high-level view of unfair practices** in crypto and DeFi, bridging technical analysis with policy-relevant insights without requiring full transaction-level enforcement.

3.3.5. Regulatory impact

CIPHER could be leveraged to benchmark crypto regulatory frameworks **across jurisdictions**[78], helping to assess risks of **regulatory arbitrage**[79] and potential cross-border spillovers. This functionality is currently not implemented within the framework but would be technically feasible if there is institutional interest.

Implementation would require **linking policy changes to on-chain behavioural shifts**, for example observing the relocation of liquidity between exchanges or changes in transaction patterns following new regulations.

Some analysis is already possible through case studies, such as e.g. examining stablecoin transaction flows[80] after regulatory interventions in major jurisdictions.

3.3.6. Potential transaction-level support

Although not within its current focus, the CIPHER framework could, in principle, be extended to support **supervisory** and **enforcement** activities by lawfully tracing **transactions**[81] relevant to **compliance** monitoring, **market abuse** detection, and **anti-money laundering** efforts.

Such **transaction-level monitoring** is beyond the core scope of CIPHER, which is designed to provide high-level, aggregated insights into overall market liquidity, leverage, and interconnections

[78] Cardozo, P. (2024). International Monetary fund. On Cross-Border Crypto Flows: Measurement Drivers and Policy Implications. https://www.imf.org/en/Publications/WP/Issues/2024/12/20/On-Cross-Border-Crypto-Flows-Measurement-Drivers-and-Policy-Implications-559166

[79] Bank for International Settlements. (2024). Considerations for the use of stablecoin arrangements in cross-border payments. *CPMI Papers*. https://www.bis.org/cpmi/publ/d220.htm

[80] Financial Stability Board. (2024). Cross-border Regulatory and Supervisory Issues of Global Stablecoin Arrangements in EMDEs. https://www.fsb.org/uploads/P230724.pdf

[81] There are currently various private companies focused on the task of blockchain analytics, such as Chainalysis, TMR Labs, Elliptic, or CipherTrace. These companies specialize in blockchain analysis and transaction monitoring. They provide tools for tracing transactions, identifying illicit activities, and supporting regulatory compliance in the cryptocurrency space.

rather than monitoring individual transfers and wallets. Nevertheless, the data infrastructure underlying CIPHER provides the **necessary foundation** for this type of granular analysis.

With interest from policymakers, a **dedicated branch of CIPHER could be developed to enable transaction-level monitoring**, including the analysis of wallet activity, identification of abnormal wallet-specific patterns such as high transaction costs or block delays, and tracing potentially illicit flows in respect of the law and in the context of lawful investigations, while complementing the project's primary focus on market-wide and systemic indicators.

3.4. Lessons learned

The development of CIPHER has highlighted a range of **challenges** that inform both its ongoing refinement and potential operational deployment.

Managing the continuous growth of blockchain data presents significant complexities, requiring careful design for **efficient storage**, **indexing**, and **retrieval**. At the same time, the high computational requirements of the framework demand robust hardware to support timely data processing, node maintenance, and analytics, as delays on less capable systems can substantially impede near real-time monitoring.

Extracting policy-relevant insights, particularly regarding **entity attribution** and **geolocation**, remains technically demanding. For instance, the Bitcoin protocol's most used client - Bitcoin Core[82] - introduces additional noise through randomized node synchronization times. This deliberate security feature greatly **diminishes efforts at geo-localization**.

Ensuring proper functioning of the complex data ingestion, storage and analytics pipeline as well as the interoperability across multiple blockchains and data formats further underscores the need for a **sustainable technical infrastructure**. Practices such as code **version control** and detailed **documentation** are essential not only for reproducibility and maintainability, but also to **enable the framework to evolve** alongside the blockchain ecosystem. These considerations are necessary for a resilient, adaptable monitoring system capable of supporting policy-relevant analysis over the long term.

3.5. Next steps

Looking forward, several priorities have been identified to enhance CIPHER's functionality and operational relevance.

Expanding **computational capacity** and integrating a **broader range of data sources** will be essential to manage the increasing volume and complexity of blockchain activity, while improvements in indexing, storage, and processing pipelines will support more reliable and comprehensive monitoring.

Planned **feature developments** aim to extend the multi-currency coverage, improve wallet clustering and provide additional visualization dashboards. **Analytical capabilities** will also be strengthened, including a time-zone analysis attempted at geo-locating the time zones of

[82] Bitcoin Core is used by over 82% of all public Bitcoin nodes: https://coin.dance/nodes/share

transaction origins based on behavioural patterns[83] and stress testing and simulation of DeFi protocols to generate forward-looking insights into potential risk propagation.

Further **collaboration** with EU institutions and international partners is expected to facilitate shared analyses, benchmarking, and convergence on monitoring standards, thereby reinforcing the evidence base for policy and regulatory decision-making.

Ultimately, the goal is to establish a **fully operational framework** capable of automated monitoring, analytics, and reporting, designed for maintainability, scalability, and adaptability to support long-term policy design, supervisory oversight, and strategic decision-making.

[83] Tubino, R. R. (2025). Temporal and Geographical Analysis of Real Economic Activities in the Bitcoin Blockchain. *Springer nature conference paper*. https://link.springer.com/chapter/10.1007/978-3-031-74643-7_17

4. Conclusions

The development of CIPHER has **demonstrated the feasibility and added value** of a systematic framework for **monitoring crypto-asset markets**. By integrating blockchain data with analytical tools tailored to policy needs, the project has shown how dimensions such as liquidity, leverage, market interconnections, and stablecoin dynamics can be transformed into policy-relevant indicators. Early use cases highlight both the potential of CIPHER to deliver timely insights and the technical challenges that must be addressed to ensure **robustness, scalability, and sustainability**.

While the current focus remains on providing **high-level, aggregate views**, the underlying data infrastructure opens pathways for **extensions into stress testing, regulatory benchmarking, or even more granular monitoring**. The flexibility of the framework ensures that CIPHER can evolve alongside the fast-changing crypto ecosystem, maintaining relevance for financial stability analysis, strategic autonomy assessments, and the evaluation of unfair market practices.

As the project advances, the priority will be to consolidate its **technical foundations**, and align outputs with concrete **policy workflows**. In doing so, CIPHER can strengthen the evidence base for regulation and supervision, support early detection of emerging risks, and enhance the EU's capacity to respond to developments in digital finance with timely, data-driven policy action.

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List of abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
AML	Anti-Money Laundering
AMM	Automatic Market Maker
API	Application Programming Interface
BIS	Bank for International Settlements
BTC	Bitcoin
CASP	Crypto Asset Service Provider
CBDC	Central Bank Digital Currency
CEX	Centralized Exchange
CIPHER	Crypto Intelligence Platform for a Holistic European Regulation
CFT	Countering the Financing of Terrorism
DAO	Decentralized Autonomous Organization
DeFi	Decentralized Finance
DLT	Distributed Ledger Technology
EBA	European Banking Authority
ECB	European Central Bank
ETF	Exchange-Traded Fund
ETL	Extract, Transform, Load
ESMA	European Securities and Markets Authority
FATF	Financial Action Task Force
FSB	Financial Stability Board

Abbreviation	Definition
JRC	Joint Research Centre
L2	Layer-2
MEV	Maximal Extractable Value
MiCA	Markets in Crypto Assets
RWA	Real World Asset
SVB	Silicon Valley Bank
TradFi	Traditional Finance
USDC	USD Coin
Zk	Zero knowledge

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Annexes

Annex 1. Data pipeline

The CIPHER platform is built with modularity and scalability in mind, separating concerns across ingestion, storage, analytics, and visualization layers. This modular design facilitates the integration of new data sources, blockchains, and analytical models with minimal disruption.

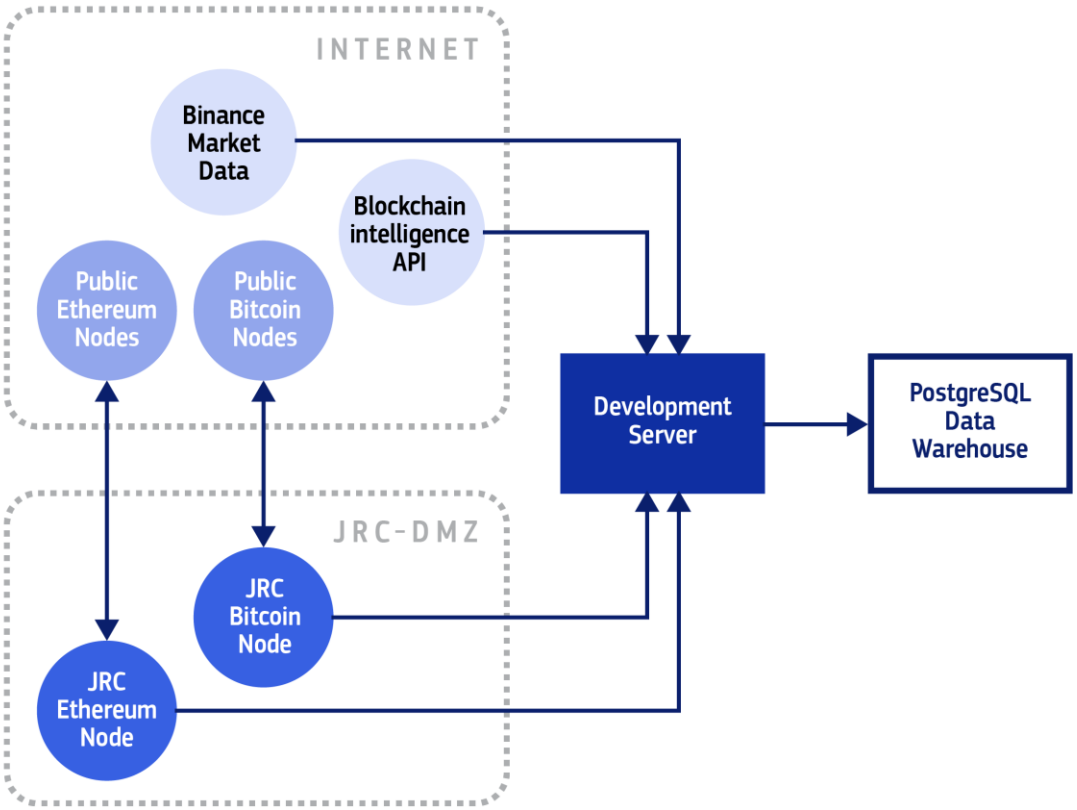
At the core of the infrastructure, two dedicated **blockchain nodes** operate within a demilitarized zone (DMZ), one for Bitcoin and one for Ethereum. These nodes maintain direct connections to public blockchain networks, ensuring reliable access to the latest ledger data while minimizing exposure of internal systems to the public internet.

A central **development server** interfaces with these blockchain nodes to ingest on-chain data and perform preliminary processing. In addition to the blockchain feeds, the development server retrieves market price data from Binance and receives entity attribution data via API from an external provider. This consolidated ingestion process allows the platform to harmonize multiple data streams for downstream analysis.

The processed data is then transferred to a dedicated **data warehouse** for long-term storage and advanced analytics, including a dashboard used for all high-level visualizations.

For a full overview of the pipeline, see **Figure 6**.

Figure 6. Individual components of the data pipeline



Source: own visualization

Annex 2. Database schema

The database of the CIPHER project currently contains two schemas, one for Bitcoin and one for Ethereum.

In the case of the Bitcoin blockchain, the main table **block** contains basic information about each block in the blockchain, such as its number, hash, etc. Paired to this table via the block number is the **transaction** table, which contains details about individual transactions such as their index, hash, fee etc. Additional variables as well as the pairing of inputs to outputs are stored in three additional tables, see **Figure 7** for further details.[84]

Figure 7. Database schema for the Bitcoin blockchain

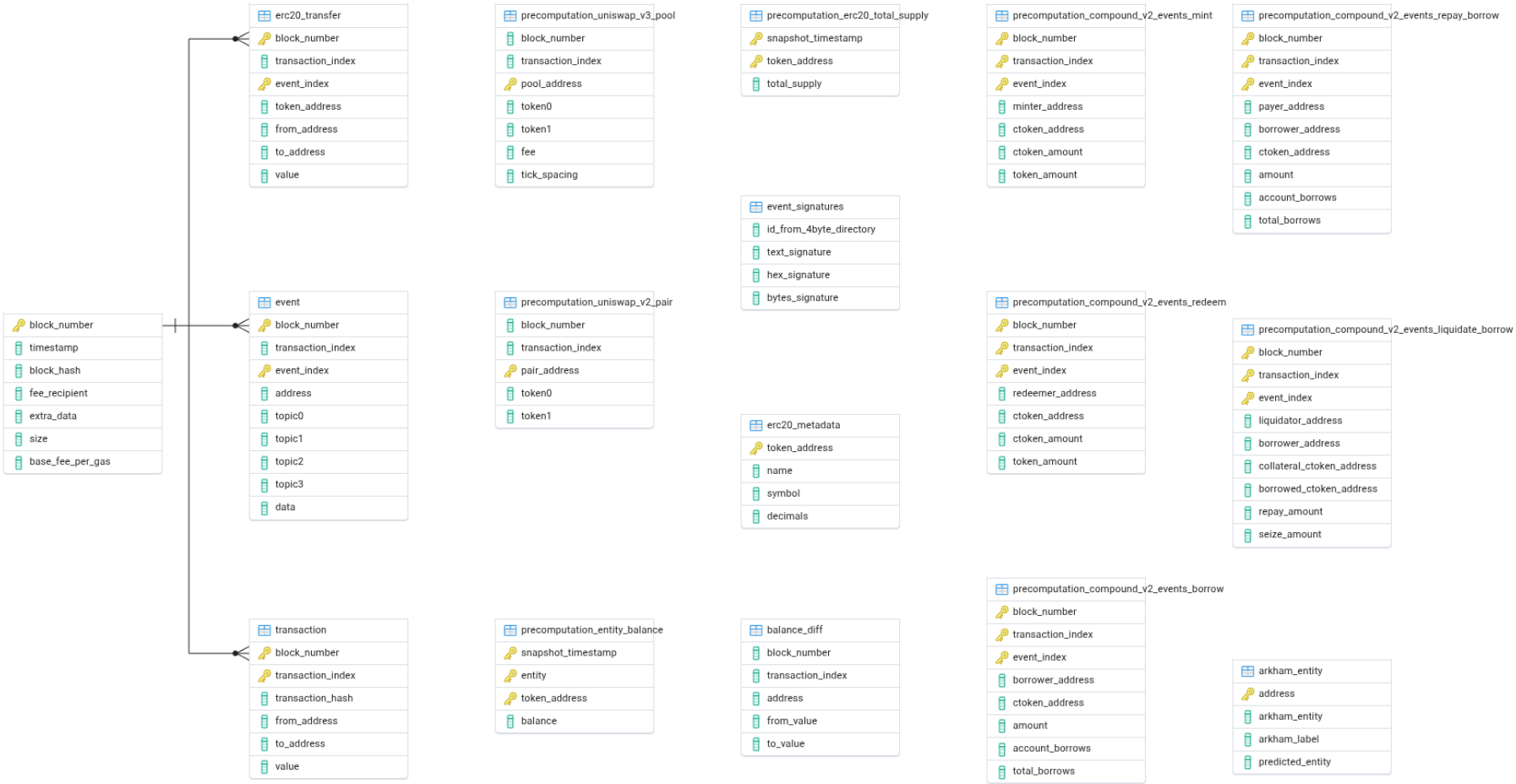


Source: own visualization via pgAdmin

For the Ethereum blockchain, the main **block** table is connected to tables **erc20_transfer**, **transaction**, and **event**. The former two store information on the transfers of ERC 20 tokens and individual transactions, while the latter contains logs of all events happening at the Ethereum blockchain. These are then useful for other tables in the schema, which allow saving of important events for specific protocols (currently implemented for Uniswap and Compound, with additional protocols pending to be added soon). Additional tables in the schema are used for precomputation of balances (used in the visualization layer) or storage of tagged entity wallets. For a complete overview of the schema, see **Figure 8**.

[84] Additional foreign keys might be added into the schema in the future for greater transparency.

Figure 8. Database schema for the Ethereum blockchain

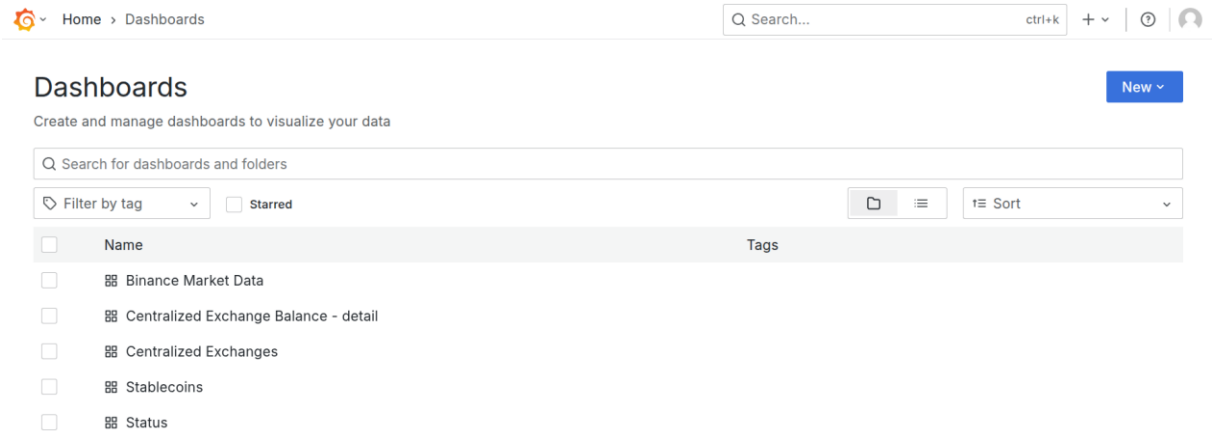


Source: own visualization via pgAdmin

Annex 3. Visualization dashboards

The CIPHER framework includes a browser-based visualization layer built on Grafana, designed to provide policymakers and analysts with intuitive access to blockchain-derived indicators. This interface currently supports four main categories of dashboards, focused on token prices, exchange balances, stablecoin volume and synchronization status. See **Figure 9** for a screenshot of the dashboard selection tool.

Figure 9. The browser-based visualization platform, showing the different dashboards available within the CIPHER framework.



Source: own screenshot of the CIPHER visualization platform

The Binance **token price dashboard** enables the monitoring of BTC/USDT and ETH/USDT pairs, which serve as primary links between the crypto-asset ecosystem and the U.S. dollar. These prices are updated in near real-time and displayed as time series, supporting both short-term monitoring and long-term trend analysis.

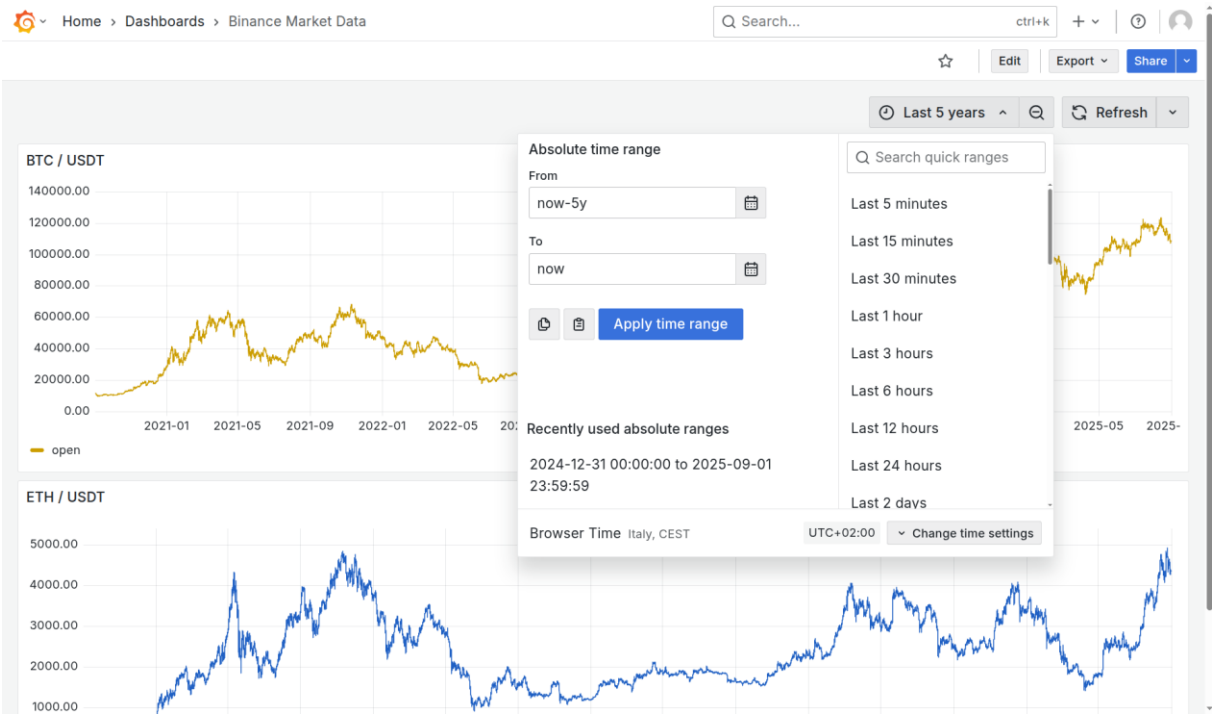
Two **exchange balance dashboards** provide insights into the reserves held on major centralized platforms, currently covering Binance, Bybit, OKX, and Coinbase, with a focus on USDT and USDC holdings. These visualizations highlight concentration risks and liquidity dynamics across venues, and they are designed to be extensible to additional exchanges and tokens as needed.

A dedicated **stablecoin volume dashboard** tracks adoption trends across major stablecoins, including USDC, USDT, USDS, USDe, and DAI. This tool supports the analysis of regulatory and policy impacts on transaction activity, as demonstrated by the sharp increase in USDe volumes following the adoption of the Genius Act (see **Figure 4**).

Finally, the **synchronization status dashboard** provides a technical overview of data ingestion, showing the state of node synchronization and data pipeline health, thereby ensuring that users can assess the reliability of the displayed indicators.

All time-series dashboards support dynamic zooming, allowing users to move seamlessly from long-horizon views down to individual block-level data. This capability ensures that the system can be used both for high-level monitoring and for more granular explorations of market dynamics. Screenshot of this functionality can be found in **Figure 10**.

Figure 10. The browser-based visualization framework, showing the crypto token price dashboard with the ability to select a specific time range.



Source: own screenshot of the CIPHER visualization platform

Annex 4. Glossary

Concept	Definition
Aave	A decentralized finance (DeFi) lending protocol that allows users to deposit digital assets to earn interest and to borrow against posted collateral. It supports overcollateralized loans and features such as flash loans and interest rate switching.
Address	A unique identifier on a blockchain representing the destination or source of cryptocurrency transactions. Addresses are derived from public keys and can be used to send, receive, and store tokens.
Algorithmic stablecoin	A type of stablecoin that maintains its peg to a reference asset (like the US dollar) through algorithmic mechanisms and smart contracts rather than being fully backed by reserves. Its value stability relies on incentives, supply adjustments, or collateralized protocols.
Arbitrum	A Layer-2 scaling solution for Ethereum that uses optimistic rollups to process transactions more efficiently and at lower cost. It bundles multiple transactions off-chain before settling them on the Ethereum mainnet, thereby increasing throughput.
Asset-referenced token	A token whose value is linked to one or more underlying assets, such as fiat currencies, commodities, or baskets of assets. These tokens aim to provide price stability or exposure to traditional assets in a digital form.
Automated Market Maker (AMM)	A decentralized trading protocol that uses smart contracts to create liquidity pools and automatically determine token prices based on supply and demand, eliminating the need for traditional order books. Commonly used in decentralized exchanges (DEXs).
BuilderNet	A specialized block builder in the Ethereum ecosystem that constructs transaction blocks for validators under the Proposer–Builder Separation (PBS) framework. Builders like Beaverbuild compete to maximize block value, often by organizing transactions to capture arbitrage opportunities.
Binance	A major centralized cryptocurrency exchange offering trading, staking, and other crypto services. Binance provides access to spot and derivatives markets and has been involved in regulatory scrutiny across multiple jurisdictions.

Concept	Definition
Bitcoin client	Software that allows a user to interact with the Bitcoin network, validating transactions, maintaining a copy of the blockchain, and sending/receiving bitcoins. Examples include Bitcoin Core and lightweight clients that rely on external nodes.
Block	A data structure in a blockchain that groups together a set of validated transactions. Each block references the previous one, creating a chronological chain that secures transaction history through cryptographic linkage.
Blockchain	A distributed, immutable ledger that records transactions in a series of linked blocks. Each block contains a set of transactions and a cryptographic hash of the previous block, ensuring transparency, security, and resistance to tampering.
bZx flash loan attack	A 2020 exploit of the bZx lending protocol that used flash loans to manipulate on-chain prices and extract profit. It was one of the first major incidents to highlight vulnerabilities in decentralized finance protocols and the systemic risks of price oracle manipulation.
Central Bank Digital Currency (CBDC)	A digital form of a country's fiat currency issued and regulated by its central bank. CBDCs aim to provide a secure, efficient, and accessible means of payment, combining aspects of traditional money with digital technology.
Centralized exchange	A cryptocurrency trading platform operated by a centralized organization that holds users' funds and manages order matching. CEXs provide liquidity and user-friendly interfaces but require trust in the operator for security and custody.
Chainlink	A decentralized oracle network that provides blockchains with external data such as asset prices, weather information, or other off-chain inputs. It is widely used in decentralized finance to secure smart contracts that depend on real-world data.
CLARITY Act	A U.S. federal bill introduced on July 20, 2023. It establishes a clear regulatory framework for the issuance and oversight of payment stablecoins - digital assets redeemable at a fixed monetary value and designed for use as a means of payment.

Concept	Definition
Compound	A decentralized finance (DeFi) protocol that enables users to lend and borrow digital assets. Interest rates are algorithmically set based on supply and demand, and participants receive governance tokens (COMP) that allow them to vote on protocol changes.
Cross-chain bridge	A technology that allows the transfer of crypto assets or data between different blockchain networks. Bridges can be trusted (centralized) or trustless (decentralized), and they typically lock assets on the source chain and mint equivalent tokens on the destination chain. They are critical for interoperability but can introduce security risks.
Crypto asset	A digital representation of value or rights that is stored on a blockchain or another DLT. Crypto assets include cryptocurrencies (like Bitcoin), stablecoins, tokenized assets, and utility or governance tokens. They may be used for payment, investment, or access to decentralized applications.
Crypto Asset Service Provider (CASP)	An entity that offers one or more services related to crypto assets, such as custody, exchange, trading, issuance, or transfer. CASPs are typically subject to anti-money laundering (AML) and counter-terrorist financing (CFT) regulations in many jurisdictions, including under EU rules like the 5th and 6th AML Directives.
DAI	A decentralized stablecoin issued by MakerDAO, collateralized through over-collateralized crypto assets such as ETH and USDC. Its peg to the U.S. dollar is maintained by smart contracts and governance mechanisms, making it one of the most established and widely used decentralized stablecoins.
Decentralized Autonomous Organizations (DAOs)	Member-governed entity built on blockchain smart contracts that allows decisions, such as protocol upgrades or fund allocations, to be made through token-based voting mechanisms. DAOs seek to replace centralized management structures with decentralized governance.
Decentralized Finance (DeFi)	A financial ecosystem built on blockchain networks that allows users to lend, borrow, trade, and invest without intermediaries. DeFi relies on smart contracts, transparency, and programmable protocols rather than centralized institutions.

Concept	Definition
Decentralized Governance	A system in which decisions about a blockchain protocol or DeFi platform are made collectively by token holders or stakeholders, often through voting mechanisms, rather than by a central authority. This promotes distributed control, though decision-making power may still vary with token holdings.
Distributed Ledger Technology (DLT)	A digital system for recording transactions across multiple locations or participants, where entries are immutable and synchronized without a central authority. Blockchain is the most widely known form of DLT.
DORA	An EU regulation establishing requirements for financial entities to strengthen their operational resilience, particularly against ICT-related risks, including those arising from the adoption of digital and crypto technologies.
Entity	Any identifiable participant in the crypto ecosystem that hold, transfer, or manage crypto assets. Entities are relevant for regulatory, compliance, and monitoring purposes, such as AML/CFT reporting and market surveillance.
ERC-20 Token	A standard type of fungible token on the Ethereum blockchain. It defines a set of rules that all compliant tokens must follow, which allows them to interact seamlessly with wallets, exchanges, and smart contracts.
Ethereum	A decentralized blockchain platform that enables the creation of smart contracts and decentralized applications (dApps). Its native cryptocurrency, Ether (ETH), is used to pay for transaction fees and computational services on the network.
Flash loan	A type of uncollateralized loan in decentralized finance (DeFi) that must be borrowed and repaid within a single blockchain transaction. Flash loans are often used for arbitrage, collateral swaps, or exploit strategies.
FTX	A centralized cryptocurrency exchange that offered trading, derivatives, and other crypto services. FTX collapsed in November 2022, highlighting risks related to leverage, mismanagement, and lack of regulatory oversight in crypto markets.

Concept	Definition
Gas	A fee paid on blockchain networks like Ethereum to execute transactions or run smart contracts. Gas compensates validators for computational work and prevents network spam.
GENIUS Act	A U.S. federal law enacted on July 18, 2025, focusing specifically on the regulation of payment stablecoins. It establishes a comprehensive framework requiring stablecoins to be backed one-for-one by U.S. dollars or other low-risk assets, mandates audits, and enforces transparency for issuers. The act also subjects stablecoin issuers to the Bank Secrecy Act, obligating them to implement anti-money laundering and sanctions compliance programs.
Harmony bridge breach	A 2022 security incident in which the Harmony blockchain's cross-chain bridge was exploited, resulting in the theft of approximately $100 million in cryptocurrency. It highlighted vulnerabilities in cross-chain protocols and smart contract security.
Layer-1	The foundational blockchain protocol that operates independently and provides core functions such as consensus, security, and transaction validation. Examples include Bitcoin, Ethereum, and Solana. Layer-1 blockchains are responsible for maintaining the integrity of the ledger and executing smart contracts natively, but they often face limitations in scalability, transaction throughput, and latency.
Layer-2	A secondary protocol built on top of a Layer-1 blockchain to improve scalability, transaction speed, and reduce costs while inheriting the security of the underlying chain. Layer-2 solutions, such as state channels, rollups, and sidechains, process transactions off-chain or in aggregated form and then settle them on Layer-1. The design of Layer-2 solutions balances higher throughput and lower fees with trust assumptions: some rely entirely on Layer-1 security, maintaining decentralization, while others introduce varying degrees of trust in operators or validators. Properly designed Layer-2 solutions preserve decentralization and trustlessness to the extent allowed by the underlying Layer-1 protocol.

Concept	Definition
Lido	A liquid staking protocol that allows users to stake cryptocurrencies (such as Ethereum) without locking their assets or managing validator nodes directly. In return, users receive tokenized staked assets (e.g., stETH) that can be used in other DeFi applications, enabling liquidity while earning staking rewards.
Loopring	A Layer-2 protocol, and decentralized exchange (DEX) protocol built on Ethereum, using zkRollup technology to increase transaction throughput and reduce fees while retaining the security of the Ethereum blockchain. Loopring supports trading, payments, and NFTs with minimal gas costs.
Mainnet	The primary blockchain network where actual transactions are executed and recorded, as opposed to testnets, which are used for development and experimentation. Mainnets carry real economic value and enforce the protocol's consensus rules.
Memecoin	A cryptocurrency created primarily as a joke, meme, or social phenomenon rather than for fundamental utility or technological innovation. Popular examples include Dogecoin and Shiba Inu. Memecoins often experience extreme price volatility driven by online communities, social media hype, and speculative trading.
MEV	The profit that miners, validators, or sequencers can earn by reordering, including, or censoring transactions within a block. MEV strategies include front-running, back-running, and sandwich attacks, which can affect transaction fairness and market integrity in DeFi.
MiCA	A European Union regulation establishing a comprehensive legal framework for crypto-assets, issuers, and service providers. MiCA provides rules on transparency, investor protection, and operational requirements to enhance market integrity and legal certainty across the EU.
Minting	The process of creating new tokens or coins on a blockchain. Minting can occur through mining (Proof-of-Work), staking (Proof-of-Stake), or other protocol-specific mechanisms, and is often used to introduce new supply or reward participants in a network.

Concept	Definition
Monero	A privacy-focused cryptocurrency that uses advanced cryptographic techniques, such as ring signatures and stealth addresses, to obfuscate transaction details. Monero ensures that sender, receiver, and transaction amounts remain confidential, prioritizing fungibility and user privacy.
Node	A participant in a blockchain network that maintains a copy of the ledger. Depending on its type, a node may validate transactions, propagate data, or simply observe the network. Full nodes typically validate and relay transactions, while light nodes rely on full nodes for verification.
Nomad hack	A major 2022 crypto exploit targeting the Nomad bridge, where attackers drained hundreds of millions of dollars by exploiting a vulnerability in the smart contract that allowed unauthorized withdrawals.
Off-chain	Refers to activities, transactions, or data that occur outside the blockchain. Off-chain operations can improve scalability and speed but may rely on trust in third parties rather than the decentralized network.
Optimism	A Layer-2 scaling solution for Ethereum that uses optimistic rollups to increase transaction throughput and reduce fees while inheriting Ethereum's security guarantees.
Optimistic rollup	A Layer-2 protocol that executes transactions off-chain and posts transaction data to the Ethereum mainnet, assuming transactions are valid ("optimistically") but allowing for fraud proofs if disputes arise. It balances scalability, cost efficiency, and decentralization.
Oracle	A service or system that provides external real-world data to a blockchain, enabling smart contracts to execute based on information such as prices, weather, or events that exist outside the chain.
Peg / de-peg	A peg is when one asset is fixed to another's value (e.g., 1 stablecoin = 1 USD). A de-peg is when it no longer holds that value.

Concept	Definition
Poly network hack	A major DeFi exploit in August 2021 in which attackers drained over $600 million from the Poly Network protocol, highlighting vulnerabilities in cross-chain interoperability and smart contract security.
Pseudonymity	A characteristic of many blockchain systems where users interact using cryptographic addresses or identifiers instead of real-world identities, providing privacy while still allowing transactions to be publicly traceable.
Pump-and-dump	A market manipulation strategy where the price of an asset is artificially inflated (“pumped”) through coordinated buying or misleading information, then sold off (“dumped”) by insiders for profit, often harming unsuspecting investors.
Qubit exploit	A security breach targeting the Qubit DeFi protocol. Attackers exploited a vulnerability in Qubit's cross-chain bridge, QBridge, to mint an unlimited amount of wrapped Ethereum on Binance Smart Chain without depositing any Ethereum on the Ethereum network. The exploit was facilitated by a logical flaw in the smart contract's deposit function, which failed to properly validate the transaction inputs.
Redemption	The process by which a holder of a token or stablecoin exchanges it for another asset, typically its underlying collateral or equivalent value. During redemption, the token is destroyed, or "burned", to remove it from circulation. For instance, a USDT token can be redeemed for dollars, while cUSDT (a token native to the Compound DeFi protocol) can be redeemed for USDT.
Ronin attack	A major 2022 security breach of the Ronin Network, the sidechain supporting the Axie Infinity game. Attackers exploited compromised private keys to steal 173,600 ETH and 25.5 million USDC (around $620 million at the time). The hack highlighted vulnerabilities in validator security and the risks of centralized control in nominally decentralized networks.
RWA Tokenization	The process of representing real-world assets (RWAs), such as real estate, bonds, or commodities, on a blockchain as digital tokens. This allows fractional ownership, easier transfer, and enhanced liquidity while maintaining a link to the underlying asset.

Concept	Definition
Smart contract	A self-executing program stored on a blockchain that automatically enforces the rules of an agreement when predefined conditions are met, eliminating the need for intermediaries.
Stablecoin	A type of cryptocurrency designed to maintain a stable value relative to a reference asset, such as a fiat currency (e.g., USD) or a commodity, often used for payments, trading, or as a store of value in crypto markets.
Staking	The process of locking up a cryptocurrency in a blockchain network to support operations like transaction validation or consensus. Participants often earn rewards proportional to the amount and duration of their stake.
StarkNet	A Layer-2 scaling solution for Ethereum using zero-knowledge proofs (ZK-rollups) to increase transaction throughput and reduce fees, while inheriting Ethereum's security and decentralization.
Terra / Luna	A blockchain ecosystem built around the Terra stablecoin (e.g., UST) and its native token, LUNA. LUNA was used to stabilize Terra stablecoins and participate in network governance. The collapse of Terra in May 2022 demonstrated systemic risks in algorithmic stablecoin designs.
Titan Builder	A neutral Ethereum block builder specializing in maximizing transaction inclusion value by aggregating and submitting bundles of transactions. The platform employs multiple sorting algorithms to optimize block construction, considering factors beyond effective gas price to enhance overall block value.
Token	A digital asset issued on a blockchain that can represent a wide variety of rights or value, such as ownership, access to a service, or participation in governance. Tokens can be fungible (interchangeable, like cryptocurrencies) or non-fungible (unique, like NFTs).

Concept	Definition
Tornado Cash	A privacy-focused Ethereum mixer that enables users to obfuscate the origin and destination of transactions. It uses smart contracts to break the on-chain link between sending and receiving addresses. Tornado Cash has faced regulatory scrutiny and sanctions due to its use in money laundering and illicit activity.
Total Value Locked (TVL)	The total value of assets deposited in a DeFi protocol, often used as a measure of its size, adoption, and liquidity.
TradFi	Short for “Traditional Finance,” referring to conventional financial systems, institutions, and markets, such as banks, stock exchanges, and regulatory frameworks, as opposed to decentralized finance (DeFi) built on blockchain technology.
Uniswap	A decentralized exchange (DEX) on Ethereum that allows users to swap ERC-20 tokens directly from their wallets without relying on a centralized intermediary. Uniswap uses automated market makers (AMMs) instead of traditional order books, relying on liquidity pools funded by users who earn fees in return.
USDC	A transparent stablecoin issued by Circle, a regulated financial institution. It is 100% backed by cash and cash-equivalent assets, with regularly audited reserves. Circle actively collaborates with regulators and has obtained necessary licenses to maintain compliance and transparency in its operations.
USDe	An algorithmic stablecoin issued by Ethena, backed not by fiat reserves but by a combination of staked Ethereum and delta-hedged derivatives positions. Its design aims to provide a synthetic dollar that is censorship-resistant while still maintaining stability, and it has grown rapidly following recent regulatory and market developments.
USDT	A stablecoin issued by Tether, which has faced criticism for its lack of transparency and regulatory compliance. Tether has not undergone independent audits by certified third-party auditors, and its reserve holdings have been a subject of scrutiny. Tether's operations have also been linked to El Salvador, where it has faced ethical and regulatory concerns due to its involvement with the country's government and its relocation to the region for tax and regulatory leniency.

Concept	Definition
Wallet	A software or hardware tool that allows users to store, send, and receive cryptocurrencies or tokens. Wallets can be custodial (managed by a third party) or non-custodial (fully controlled by the user).
Wash trading	A market manipulation practice where an entity simultaneously buys and sells the same asset to create artificial trading volume, misleading other participants about demand or liquidity.
Wormhole exploit	A major 2022 security breach of the Wormhole cross-chain bridge, where attackers exploited a vulnerability in the bridge's smart contract to mint 120,000 wrapped Ethereum (wETH) without collateral, resulting in a loss of around $320 million. The incident highlighted the risks of bridging assets between blockchains and vulnerabilities in smart contract design.
Wrapped token	A tokenized version of a cryptocurrency that exists on a blockchain other than its native chain, allowing it to be used in DeFi applications. For example, Wrapped Bitcoin (WBTC) is an ERC-20 token on Ethereum representing Bitcoin.
Zcash	A privacy-focused cryptocurrency that uses zero-knowledge proofs (zk-SNARKs) to enable shielded transactions, hiding the sender, receiver, and transaction amount on its blockchain.
Zero knowledge rollup (zkRollup)	A layer-2 scaling solution for blockchains that bundles multiple transactions off-chain and submits a cryptographic proof (zero-knowledge proof) on-chain, ensuring correctness while reducing fees and improving throughput.
zkSync	A layer-2 scaling solution for Ethereum using zero-knowledge rollups (zkRollups) to increase transaction throughput and reduce fees while inheriting Ethereum's security.

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